Kernel Affine Hull Features for Prediction: Conditional Gains, Costs, and Limits of Transfer*

Anonymous authors
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Abstract

Kernel Affine Hull Machines (KAHMs) learn local reconstruction geometry, but reconstruction quality alone does not determine predictive usefulness. We investigate the roles of retained position, decoder capacity, computational cost, and transfer using controlled experiments and a separate natural-source forecasting study. On a fixed branch distribution, changing the target reveals a positional distinction that matched affine scores nearly remove and KAHM scores retain. Locality-aware affine scores and explicit local coordinates largely repair the deficit, so the mechanism is not exclusive to KAHM. A fresh-sample study with standardized geometry finds KAHM local-coordinate gains on Van der Pol at two decoder capacities, alongside adverse comparisons on autoregression and Duffing. A subsequent evaluation uses eight natural-data source families, with four reserved for confirmation after freezing the model-selection diagnostic. On those sources, raw-history-plus-KAHM has 4.26% higher source-balanced scaled mean-squared error than the matched raw-history predictor at 2.30 times its measured warm-library pipeline cost. Adding geometric features to metadata and a paid validation probe changes none of 80 confirmation choices. A subsequent paired, post-hoc min–max sensitivity leaves the natural error gap almost unchanged, at 4.29% above the matched raw predictor. It changes individual controlled errors without reversing their main mean rankings. These results support evaluating geometric features together with their decoders, structural controls, and selection costs. They establish conditional benefits and bounded failures of transfer, without a general KAHM adoption criterion. The accompanying artifact preserves both studies and reconstructs the reported numerical evidence.

1 Introduction

Kernel Affine Hull Machines (KAHMs) provide nonlinear reconstruction maps from local reference observations (Kumar et al., 2024). Their use as predictive representations raises a sequence of questions: which target variation survives the representation, can a finite decoder use it, and does any resulting improvement justify the complete fitted procedure? These questions concern different objects. A local reconstructor may recover every point on an affine line exactly, while its zero residual omits position along that line. A predictor receiving only this residual cannot use the position, even though the reconstructor used it. Conversely, preserving the raw input does not ensure that a particular regularized decoder benefits from appended geometric features.

We examine this distinction in three connected but separately evaluated steps. First, a controlled input distribution varies the target's dependence on branch identity and signed local position. Matched banks separate the reconstruction rule from reference selection, principal directions, and score encoding. This intervention identifies a positional-information bottleneck and controls that repair it. Second, a fresh-sample capacity study tests the resulting local-coordinate architecture on three synthetic forecasting laws. It finds reproducible KAHM gains on Van der Pol, harms on neighboring laws, and dependence on the finite decoder.

Third, a separate natural-source study asks whether an augmented KAHM procedure is useful at its measured computational cost, and whether a frozen diagnostic anticipates its benefit on reserved sources. Its confirmation

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results under its fixed standardization are unfavorable overall: raw-history-plus-KAHM has scaled MSE 0.8419, versus 0.8075 for the matched raw-history predictor and 0.7798 for raw ridge. Adding two geometric descriptors to metadata and a validation probe does not change the diagnostic's confirmation decisions. These observations delimit the practical conclusions that can be drawn from the favorable controlled example.

A bounded follow-up then changes geometry scaling while preserving the raw input blocks, seeds, target units, capacities, and validation grids. On the same four formerly reserved sources, min–max KAHM has scaled MSE 0.8422, versus 0.8419 under standardization; the natural-data ranking persists. This new sensitivity reuses inspected data, so its paired comparisons have a different evidential status from the original frozen evaluation.

The paper contributes (i) a controlled account of how KAHM reconstruction, encoding, and local position interact with the target; (ii) a separately specified, fresh-sample demonstration of conditional finite-decoder gains; and (iii) an auditable natural-source evaluation of accuracy, complete measured pipeline dependencies, and an unsuccessful benefit diagnostic, supplemented by a bounded analysis of scaling dependence. Local tangent features, finite kernel approximation, and feature-based selection are established ideas. Our contribution is the evidence about their use with KAHM and the implications for evaluating reconstruction-based features, rather than a new universal learning principle or performance certificate.

The stages have different estimands and implementations. The controlled capacity study uses local-only representations, historical query arithmetic, and 64/256 landmarks. The natural study uses stable query evaluation, raw augmentation in its primary pair, 32/64 landmarks, independently fitted alternatives, and its own data and tuning limits. It tests practical reach under a new protocol; it is not an identical-parameter replication of the Van der Pol effect. We keep their statistical analyses and conclusions separate throughout.

2 Related work

Reconstruction geometry and local prediction. KAHMs were introduced for geometrically informed classification, including a differential-privacy application (Kumar et al., 2024), and developed for collaborative learning (Kumar et al., 2025). Earlier geometric classification uses affine feature lines (Li & Lu, 1999); kernel sparse representation studies reconstruction-based codes in implicit feature spaces (Gao et al., 2010). Their original task results do not establish that a reconstruction score preserves the variation required by a different predictive target. Local coordinate coding connects localized representations with approximation of smooth functions (Yu et al., 2009). Local tangent coding explicitly augments coding weights with weighted principal-direction displacements (Yu & Zhang, 2010), the feature form used here. With linear readouts this also resembles local affine experts (Jacobs et al., 1991); our gates are fitted without outcomes.

Target information and finite fitting. Supervised dimension reduction asks which features preserve the input–target relationship (Fukumizu et al., 2004). Predictive $\mathcal{V}$-information distinguishes information accessible to a restricted observer (Xu et al., 2020). Our interventions and finite-decoder comparisons make a related distinction without estimating unrestricted Bayes risk. Nyström approximation (Williams & Seeger, 2001; Rudi et al., 2015) changes both fitting and computation; equal requested landmark counts do not equate function classes, realized ranks, or costs. Retaining an affinity normalizer gives an elementary invertible reparameterization, not a new information-recovery theorem.

Forecast evaluation and selection. Feature-based forecast selection, ranking, and combination have substantial precedents (Prudêncio & Ludermir, 2004; Montero-Manso et al., 2020; Talagala et al., 2022; 2023). Landmarking uses inexpensive learners to characterize a task (Pfahringer et al., 2000); our paid validation probe follows that principle. The tested increment is whether two geometric descriptors improve a specified diagnostic beyond metadata and that probe. Model-selection bias (Cawley & Talbot, 2010) and forecast-evaluation pitfalls (Hewamalage et al., 2023) motivate chronological splits, explicit cost scopes, and separation of development from reserved original sources. The Monash archive supplies the natural datasets (Godahewa et al., 2021).

Relationship to close prediction-limit studies. Kumar & Kargaran (2026) study successor soft coordinates, norm-bounded linear prediction certificates, and Koopman closure. A related unpublished manuscript (Anonymous, 2026) develops observed-output risk certificates under declared decoder sensitivity. That manuscript also reports input-scaling ablations for earlier prediction architectures. The present scaling sensitivity concerns the distinct local-coordinate and raw-augmented pipelines evaluated here. Those studies share KAHM source machinery and some generator definitions; they already distinguish fitted accuracy from predictor-class risk. The present target intervention, encoding ablations, fresh capacity comparison, and reserved natural-source evaluation address finite fitted procedures. Their paired-MSE intervals and measured selection costs do not extend the earlier class-risk or minimum-feasible-cost certificates.

3 Geometry, predictive encoding, and decoder access

Matched reconstruction rules. Let z be a training-standardized input. The controlled study uses three k-means blocks with 2, 4, and 8 local models. Each model c has at most 24 shared reference rows V_c, their center μ_c, and a common orthonormal principal basis P_c of effective rank at most $\min(2, d-1, n_c-1)$. Scaling, partitions, references, bases, and scoring are held fixed when the reconstruction rule is changed. The two rules are

$$r_c^{\mathrm{A}}(z) = \mu_c + P_c P_c^\top (z - \mu_c), \qquad r_c^{\mathrm{K}}(z) = V_c^\top \left(\frac{k_c(z) B_c}{k_c(z) B_c \mathbf{1}} \right)^\top . \tag{1}$$

Here $k_c(z)$ is the Gaussian query-kernel row in the local covariance metric and B_c is the fitted regularized inverse; their exact construction is given in Appendix B. KAHM weights may be signed. The archived controlled experiments retain the upstream float32 references and replace an exactly zero denominator by one; a separate numerical analysis below examines Gaussian underflow. This common-bank substitution isolates a reconstruction component rather than comparing independently optimized complete systems.

What the predictor receives. For either reconstruction, the matched discrepancy combines radial and angular errors:

$$\begin{aligned} &q_c(z) = 1 - \exp\{-\|z - r_c(z)\|_2\}, \qquad b_c(z) = \pi^{-1} \arccos\!\left(\mathrm{clip}\!\left(\frac{\langle z, r_c(z) \rangle}{\|z\|_2 \|r_c(z)\|_2}, -1, 1 \right) \right), \\ &T_c(z) = \sqrt{\{q_c(z)^2 + b_c(z)^2\}/2}, \qquad w_c(z) = (1 - T_c(z) + \tau)^\omega, \qquad \phi_c(z) = \frac{w_c(z)}{Z(z)}, \quad Z(z) = \sum_{j \text{ in block}} w_j(z), \end{aligned} \tag{2}$$

with $\tau = 10^{-6}$, $\omega = 16$, and cosine zero when its denominator is zero. Affinities from the three blocks are concatenated. In addition to ϕ, the ablations supply the full score T, one $\log Z$ per block, raw inputs, or gated local coordinates

$$\Gamma_{\mathrm{local}}(z) = \left[\phi(z), \{\phi_c(z) P_c^\top (z - \mu_c)\}_{c=1}^{14} \right]. \tag{3}$$

This is the established local-tangent feature form (Yu & Zhang, 2010); the investigation concerns the choice and predictive role of the gates.

These distinctions matter even before fitting. The outer normalization has the exact inverse

$$T_c = 1 + \tau - \exp\{(\log \phi_c + \log Z)/\omega\}. \tag{4}$$

Thus T and $(\phi, \log Z)$ retain the same information, without necessarily having the same fitted risk under a finite decoder. Likewise, positive gates and full column rank of the stack of $P_c^\top$ suffice to recover z from Equation (3): divide by each gate, add $P_c^\top \mu_c$, and solve the stacked linear system. Numerical full rank and positive inspected gates are observed in all 144 controlled forecasting banks; these retrospective checks do not guarantee stable inversion or population-wide gate bounds (Appendix C.5). Rank deficiency is not proof of information loss because the gates can also encode information.

A controlled target intervention. Let $S \sim \mathrm{Unif}\{-1,+1\}$, $T \sim \mathrm{Unif}[-1,1]$, and $\epsilon \sim \mathrm{Unif}[-0.1,0.1]$ be independent. Define

$$X = (T, c\cos(\pi T) + 0.5S), \qquad Y_\lambda = \sqrt{1-\lambda}\,S + \sqrt{3\lambda}\,T + \epsilon, \quad c \in \{0,1\}, \quad \lambda \in \{0, \tfrac{1}{2}, 1\}. \tag{5}$$

Within each repetition and curvature, changing λ leaves input samples, noise, and the fitted bank fixed. The signal variance is one and the Bayes MSE is $1/300$. Targets move from branch identity to signed position, while difficulty for a particular decoder may also change. A code invariant to $Q(t, x_2) = (-t, x_2)$ loses the odd conditional-mean component $\sqrt{3\lambda}T$ and has Bayes MSE at least $1/300 + \lambda$. For an approximately invariant code and a κ-Lipschitz decoder,

$$R(g \circ \Gamma) - R_X^{\mathrm{Bayes}} \geq \left(\sqrt{\lambda} - \kappa\delta_\Gamma/2\right)_+^2, \quad \delta_\Gamma^2 = \mathbb{E}[d_U(\Gamma(X), \Gamma(QX))^2]. \tag{6}$$

The measured reflection differences are descriptive; neither a population bound on δ_Γ nor a common Lipschitz budget for the fitted heads is asserted. Appendix A supplies the precise assumptions and proofs, including why small local reconstruction residuals alone are insufficient: reconstructing X from the query is different from reconstructing it through the predictor's retained code.

On straight branches with the position target, affine-affinity ridge MSE is 1.002694, versus 0.024668 for KAHM affinities, although affine reconstruction is nearly exact. The corresponding training-standardized reflection RMS is about 7.5×10^{-9} for affine affinities and 1.40 for KAHM. A locality-aware affine score reduces MSE to 0.004641; affine local coordinates attain 0.00333, close to the Bayes error. The evidence therefore identifies positional encoding as useful in this comparison, rather than an exclusive KAHM advantage. On curved branches, both codes substantially distinguish reflected positions. Full branch results, reconstruction diagnostics, and scoring controls appear in Appendix D.

4 Conditional gains in controlled forecasting

The controlled forecasting laws are nonlinear autoregression (AR), Van der Pol, and Duffing. Each repetition independently generates 128 training, 128 validation, and 512 test episodes of length 72. Histories have length 12 and predict offsets 1, 4, and 12 at 13 origins per episode. Entire episodes stay within a split; MSE averages origins and horizons within episodes and then episodes. Appendix B specifies the numerical generators and fitting rules.

The first study has 24 full data-and-fit repetitions in each of nine primary settings: six branch targets and three forecasting laws. Adding local coordinates to KAHM affinities improves ridge prediction beyond the fixed practical margin in 8 settings. On Van der Pol, mean MSE falls from 0.111108 to 0.014475, although affine local coordinates perform better with the same ridge head. The first stage uses a 29-contrast family of Bonferroni paired Student intervals. This supports retaining local position, while leaving the complete representation–decoder choice unresolved.

A separate follow-up, motivated by those outcomes, freezes its candidates, margins, and analysis before generating 24 fresh repetitions per forecasting law. It compares raw histories, affine local features, KAHM local features, and full KAHM scores with 64 or 256 Gaussian Nyström landmarks. The head appends linear coordinates to the finite nonlinear basis, standardizes basis columns, and uses an ordinary Euclidean ridge penalty. Validation selects among three bandwidth multipliers and seven penalties; the fitted model uses training observations only. Equal landmark counts do not imply equal realized ranks, function classes, or computational costs.

Table 1 retains all favorable, adverse, and unresolved comparisons. KAHM local features outperform raw and affine local features on Van der Pol at both widths in every repetition. At 256 landmarks, mean MSEs are 0.000871969, 0.001654150, and 0.001078003, respectively: descriptive reductions of 47.3% and 19.1% relative to the controls. The two simultaneous lower endpoints, 0.00075496 and 0.00018100, exceed the frozen practical margin 0.00014891; both 64-landmark comparisons also clear it. Raw inputs win on AR at both widths, and both controls win on Duffing. KAHM–affine differences on AR remain unresolved. The separate 29- and 12-contrast families do not imply a joint 95% coverage statement. Their inferential units are complete repetitions, not windows or horizons (Appendix C).

Table 1: All 12 primary fresh-sample capacity contrasts. Entries are mean MSEs and paired differences over 24 independent full repetitions, multiplied by 10^3; Δ is control minus KAHM-local loss. Intervals use the follow-up's separate 95% Bonferroni paired Student family. Coverage relies on normality or its approximation for paired differences; these are not distribution-free guarantees.

System	M	Control	Control MSE	KAHM local MSE	Δ [interval]
Nonlinear AR	64	Raw	4.4238	4.8795	−0.4556 [−0.5235, −0.3878]
Nonlinear AR	64	Affine local	4.9120	4.8795	0.0325 [−0.0031, 0.0680]
Nonlinear AR	256	Raw	4.6958	5.1749	−0.4791 [−0.6013, −0.3569]
Nonlinear AR	256	Affine local	5.2347	5.1749	0.0598 [−0.0131, 0.1327]
Van der Pol	64	Raw	2.8586	1.0266	1.8320 [1.7369, 1.9271]
Van der Pol	64	Affine local	1.3767	1.0266	0.3501 [0.2755, 0.4247]
Van der Pol	256	Raw	1.6541	0.8720	0.7822 [0.7550, 0.8094]
Van der Pol	256	Affine local	1.0780	0.8720	0.2060 [0.1810, 0.2311]
Duffing	64	Raw	3.4597	4.2154	−0.7557 [−1.0831, −0.4282]
Duffing	64	Affine local	3.0549	4.2154	−1.1605 [−1.4707, −0.8503]
Duffing	256	Raw	2.8364	4.0736	−1.2372 [−1.5242, −0.9503]
Duffing	256	Affine local	2.7981	4.0736	−1.2756 [−1.5702, −0.9810]

Numerical sensitivity and an adverse natural-data check. The original Gaussian query evaluation can produce an all-zero row before signed normalization. Replacing $\exp \ell_j$ by $\exp(\ell_j - \max_i \ell_i)$ preserves exact normalized weights whenever the denominator is nonzero (Proposition 7). In a post-hoc check reusing the 24 Van der Pol follow-up samples and banks, with affected heads refitted and validation-tuned, exactly zero normalizers fall from 99,762 among 2,236,416 query–local-model evaluations to zero. KAHM-local mean MSE changes by less than 4×10^{-13} at either width, whereas full-score losses improve by roughly 0.37–0.39%. Row scaling addresses this underflow pathway; it does not prevent signed cancellation or supply new independent confirmation.

On a fixed grouped split of Airfoil Self-Noise (Brooks et al., 1989), raw RBF obtains mean group-weighted MSE 15.6898 dB^2, compared with 20.9313 for KAHM-local RBF. The validation selector chooses raw RBF in all 24 fits. These seeds characterize fitting variability on one static-regression dataset and split, not independent natural forecasting evidence. Together, the controlled results motivate testing local features as conditional candidates. They neither explain Van der Pol through the branch symmetry nor predict transfer to new source families; Section 5 examines that question separately.

5 Natural forecasting: transfer, cost, and predicting benefit

5.1 A separate source-level evaluation

The natural study fixes four development families (Electricity, NN5, Tourism, and FRED-MD) and four confirmation families (Traffic, Hospital, Saugeen River flow, and US births). Versions, licenses, eligibility rules, and chronological windows are specified in Appendix F. Training-prefix eligibility and a fixed salted ordering select 26 series. Two history–horizon regimes, two capacities, and two seeds yield 208 runs: 128 development and 80 confirmation. Related series, windows, horizons, capacities, and seeds remain within their original source families. The units describing transfer breadth are the four confirmation families.

Each series uses chronological 60/20/20 percent training/validation/test blocks, and every target horizon lies entirely within its split. All learned transformations and feature bases use the permitted training data. Validation chooses hyperparameters; a final refit uses eligible pretest targets, and test outcomes never retune parameters. Contexts can use earlier observed values. At most 64 validation and 64 test origins are retained by a fixed spacing rule; missing windows are excluded identically for every method, without interpolation or replacement series. Mean-squared forecast error is divided by a fixed initial-training seasonal-difference scale. Aggregation gives each source family equal weight, followed by its nested task levels. These descriptive quantities are not pooled with the controlled-study confidence families.

The twelve candidates include raw ridge and raw finite-RBF predictors; affine-, locality-, and KAHM-gated local predictors, with and without the full raw history; and seasonal naive, Holt, histogram boosting, and a small MLP. The primary matched pair is raw RBF (R) versus raw plus KAHM (K). The raw-augmented code is

$$T_{\mathrm{raw}+g}(x) = \big(z(x),\, \phi_g(z(x)),\, (\phi_{g,c}(z(x))P_c^\top(z(x)-\mu_c))_c\big). \tag{7}$$

It retains every standardized history coordinate. For a fixed fitted state the raw input is therefore recoverable and Bayes information is unchanged (Proposition 8). Finite fitted performance can still change: appended features alter Gaussian distances, the Nyström basis, and ridge regularization, and the resulting nonlinear classes need not be nested.

Small/medium configurations cap training windows at 512/1,024 and RBF landmarks at 32/64. Each method fits its own preprocessing and geometry; affine/locality alternatives never fit a KAHM kernel. Within a method, features and head statistics are reused across penalties $\{0.01, 1, 100\}$. The selected model is refitted from scratch. These are specified finite procedures, not equally timed or exhaustively tuned algorithms. Detailed heads, practical controls, capacity caps, and numerical conventions appear in Appendices E–F.

5.2 Which computational work is charged?

Measured warm-library pipelines include required preparation/loading, preprocessing, validation search, selection bookkeeping, chosen refitting, forecasting, and serialization. Libraries are initialized beforehand; imports and process launch are separately exposed. Wall and CPU time, model bytes, and process high-water memory have distinct recorded scopes. Let S_m denote method m's search nodes, F_m its final fit and forecast, and m^* a validation-selected candidate. Its charged work is

$$C_{\mathrm{val}} = \sum_{v \in \mathcal{D}_{\mathrm{data}} \cup (\bigcup_{m \in \mathcal{A}} S_m) \cup F_{m^*}} c_v + c_{\mathrm{decision}}. \tag{8}$$

Unique dependency nodes are counted once. A diagnostic selector likewise pays for all required descriptors and probes, even when it rejects KAHM. Acquiring development labels and fitting the small diagnostic regressions are separate offline investments. No assumed deployment count is used to amortize them. The costs describe measured implementations on one CPU environment; they are not intrinsic complexity or hardware-independent speed ratios.

5.3 Accuracy and cost on reserved sources

All 208 declared runs completed with every candidate present; saved forecasts reconstruct all 2,496 task–method losses. Table 2 retains all twelve confirmation candidates. Raw plus KAHM has 4.26% higher aggregate error than raw RBF, at 2.30 times its measured warm-library pipeline cost. Raw ridge has lower aggregate error than both. Against raw RBF, KAHM raises mean loss by 0.50% on Traffic, 12.27% on Hospital, and 3.53% on US births; it lowers loss by 1.53% on Saugeen, where raw ridge also performs better. Leaving out any one confirmation family still leaves the KAHM aggregate above raw RBF. This sensitivity check is descriptive and does not turn four selected sources into a population-wide inference.

The structural controls clarify attribution. Raw plus KAHM improves aggregate loss over raw plus affine and raw plus locality by only 0.46% and 0.50%, respectively. Each comparison favors KAHM on two source means and the control on two; KAHM costs 14.65% and 38.82% more. These mixed differences do not identify nonlinear reconstruction as a general source of advantage. At the small/medium settings, KAHM raises loss against raw RBF by about 2.00/6.70%; the settings change both sample caps and model sizes, so their effects are not separately identified.

The favorable development mean also needs qualification. KAHM reduced loss against raw RBF by 42.45%, but FRED-MD supplied 96.62% of the net absolute reduction. Removing that family reduces the apparent gain to 4.65%. Raw plus affine had lower mean loss and cost than KAHM in each development family, and raw ridge had the lowest overall development loss. Consequently, the development comparison alone did not establish a KAHM-specific benefit.

Table 2: All twelve standalone methods on the reserved natural sources. Scaled MSE and warm-library wall time use equal source-family weights and the frozen within-source hierarchy. Times include required preprocessing, search, refitting, prediction, serialization, and measured bookkeeping; imports and launch are separate. These are descriptive results from four source families.

Method	Scaled MSE	Wall seconds per task
Raw ridge	0.7798	0.1880
Raw RBF	0.8075	0.1688
Affine local	0.9449	0.5002
Locality local	0.8964	0.2898
KAHM local	0.9354	0.3218
Raw + affine	0.8458	0.3379
Raw + locality	0.8462	0.2791
Raw + KAHM	0.8419	0.3874
Seasonal naive	1.0498	0.1614
Holt	2.3729	0.3449
Histogram boosting	0.9555	8.9103
MLP	0.8785	1.4626

Implementation overhead is substantial. Selection cleanup, including explicit garbage collection, contributes 0.1381 of raw RBF's 0.1688 seconds and 0.1825 of KAHM's 0.3874 seconds. Search and final refit are also reported separately (Table 21). The 2.30-fold ratio describes the supplied warm workflow; it is not a cold-start end-to-end deployment ratio. Figure 1 retains the full accuracy–cost comparison, including selectors that must pay to choose a model.

5.4 The frozen benefit diagnostic

Four fixed ridge diagnostics predict $\log[(\ell_R + 10^{-12})/(\ell_K + 10^{-12})]$ from metadata M, M plus geometry G, M plus a cheap validation probe P, or $M + P + G$. Metadata describe training size, horizon/history ratio, and lag-one autocorrelation; geometry adds gate entropy and an affine-projection residual. The paid probe compares reduced-capacity raw/KAHM fits on validation data. A positive estimated log benefit selects KAHM; ties select raw. Development standardization and training hold out whole source families. All variants are then fitted on development and frozen before confirmation; confirmation outcomes cannot choose the variant or threshold.

The primary geometric increment, $M + P + G$ versus $M + P$, changes none of the 80 confirmation decisions. Both achieve scaled MSE 0.807085 at about 0.3309 seconds. The probe already computes the geometric descriptors, so they need no extra bank fit in this pair. Log-benefit RMSE nonetheless increases from 0.184549 to 0.184680. The shared decision rule improves aggregate loss over always raw by only 0.054%, at 1.96 times its pipeline cost. It improves one source mean, worsens one, and ties two. Metadata-plus-geometry also makes exactly the same choices as metadata alone, while paying for extra feature acquisition. All variants and validation-portfolio controls remain in Table 22.

Offline learning requires more than fitting a small ridge regression. Acquiring the matched development outcomes and shared probe/metadata dependencies costs 80.83 seconds of summed measured pipeline work; all four final ridge fits together add about 0.032 seconds. Shared dependencies must not be added repeatedly across variants. These observations provide no validated geometric adoption criterion for the tested procedure. They also do not establish that usefulness is unpredictable by a different, independently evaluated diagnostic.

6 Post-hoc sensitivity to geometry scaling

The preceding studies fix training z-score geometry scaling. To test whether that choice determines their practical interpretation, a bounded follow-up replaces it with training min–max scaling to $[-1, 1]$, without clipping future inputs. Raw skips and references remain standardized; all targets, splits, seeds, head grids, capacities, and query conventions remain fixed. The protocol and execution code were published before

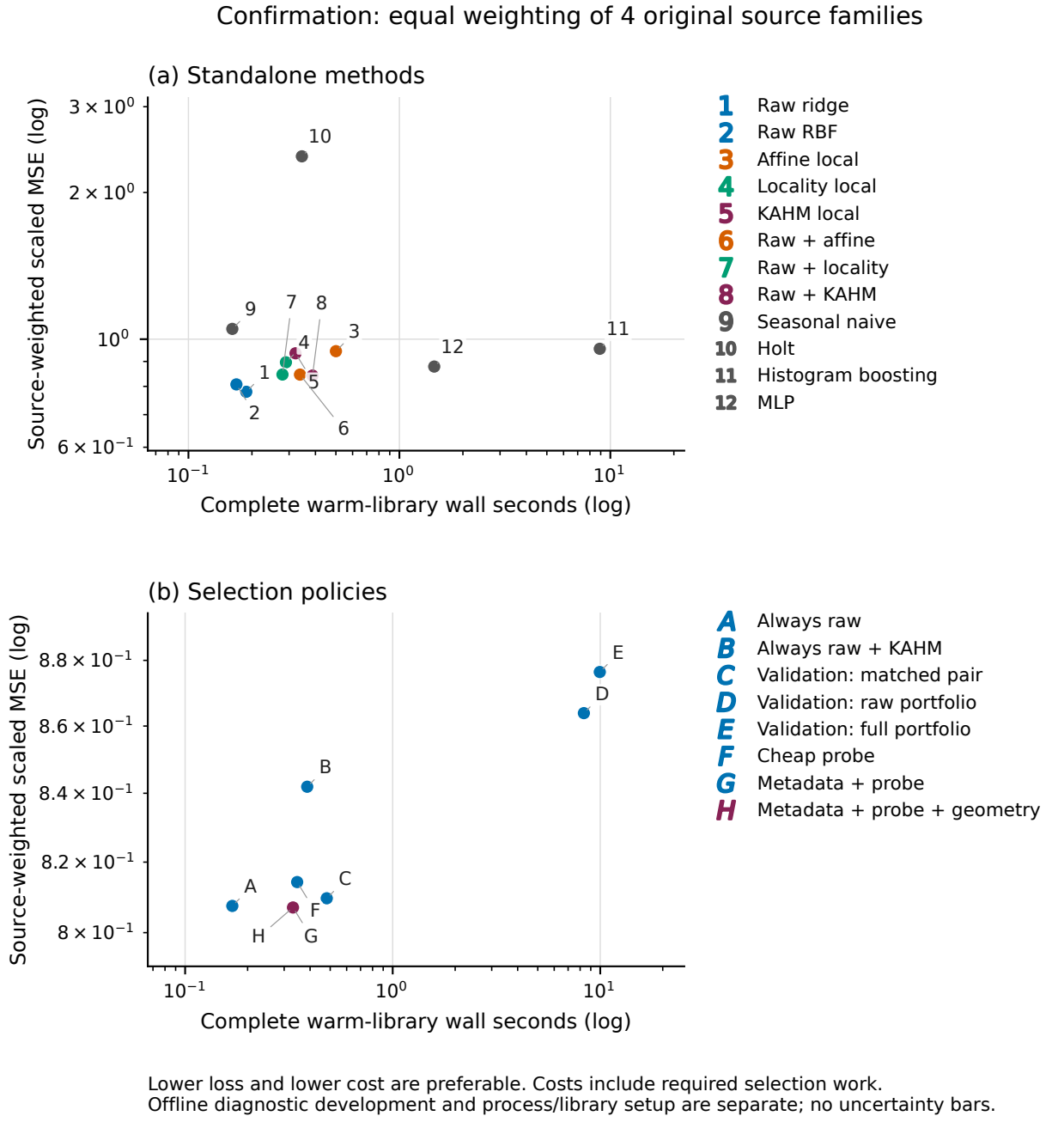


Figure 1: Reserved-source accuracy and measured warm-library cost. All points use the same source hierarchy. The standalone and selector panels have separate scales so that all declared candidates remain visible. Selector costs include required searches and probes. Confirmation outcomes are used to evaluate these procedures, not to fit a new selection rule.

the sensitivity fits, after the original outcomes had been inspected. These are paired post-hoc comparisons on reused sources and draws, not fresh confirmation. Appendix I specifies the intervention, costs, retained evidence, and complete results. The frozen benefit diagnostic is not refitted or reevaluated for min–max geometry.

Method	Standard MSE	Min–max MSE	Change (%)
Raw ridge	0.7798	0.7798	fixed
Raw RBF	0.8075	0.8075	fixed
Raw + affine	0.8458	0.8596	+1.62
Raw + locality	0.8462	0.8488	+0.31
Raw + KAHM	0.8419	0.8422	+0.03

Table 3: Geometry-scaling sensitivity on the four formerly reserved natural sources. Scaled MSE uses the original source hierarchy. Standard arms reproduce the frozen results; min–max changes geometry only, retains the raw skip, and applies no input clipping. Change is $100(\ell_{\text{minmax}}/\ell_{\text{standard}} - 1)$; negative favors min–max. Raw references are fixed. All new comparisons are post-hoc sensitivity.

Natural-source ranking. All 208 configurations are rerun for the three raw-augmented geometric methods under both scalers, with unchanged raw RBF and ridge references. On the four formerly reserved sources, KAHM's source-balanced scaled MSE changes from 0.8419 to 0.8422, an increase of 0.03% (Table 3). The min–max arm remains 4.29% above raw RBF and 8.00% above raw ridge. Min–max KAHM beats raw RBF on only Saugeen, and raw ridge has lower loss on all four source means. Leaving out any one family still leaves min–max KAHM above raw RBF. Thus this specified scaling alternative does not reverse the aggregate natural-data comparison.

The near-unchanged aggregate is not an equivalence guarantee. KAHM's loss falls by 0.15% on Traffic and 0.86% on Hospital, but rises by 0.90% on Saugeen and 0.35% on US births. Small-capacity loss increases by 0.99%, whereas medium-capacity loss falls by 0.96%; these settings still jointly vary sample and model-size caps. The original development aggregate worsens by 7.70%, with heterogeneous source effects. The same intervention also changes the affine and locality controls; it does not isolate an effect of nonlinear KAHM reconstruction.

In this new execution, KAHM's source-balanced mean min–max warm method time is 0.5561 seconds per task, versus 0.3804 for standardized KAHM and 0.1762 for the remeasured raw RBF reference: ratios of 1.46 and 3.16, respectively. These include search, selection cleanup, refit, forecast, and serialization. A retained Saugeen search/cleanup outlier contributes to the new timing aggregate. They are single-execution, implementation-specific observations, not an intrinsic computational penalty of min–max scaling or a cold-start deployment claim.

Controlled finite-decoder comparisons. All 24 original complete draws for each law and both landmark counts are reused. Both scaler arms retain raw, affine-local, KAHM-local, and KAHM-score features; a matched locality-coordinate control is explicitly added. Table 4 reports every law/capacity cell, with all representations and paired dispersion in Appendix I. The main mean-loss ordering persists: raw RBF remains better than KAHM local coordinates on autoregression and Duffing, whereas KAHM local coordinates remain better than raw and both local controls on Van der Pol at both capacities. KAHM-local scaling changes range from a 0.23% increase on Van der Pol at 256 landmarks to a 7.35% reduction on Duffing at 256 landmarks. The latter lowers MSE from 4.0736 to 3.7741 in units of 10^{-3}, but remains above raw (2.8364), min–max affine (2.7742), and min–max locality (2.6921). Scaling can materially change an error without reversing the scientific comparison. These reused-draw mean comparisons carry no new confirmatory confidence claim.

Law	M	Raw	KAHM std.	KAHM min–max	Affine min–max	Locality min–max
Nonlinear AR	64	4.4238	4.8795	4.8678	4.8940	4.9414
Nonlinear AR	256	4.6958	5.1749	5.1584	5.1807	5.2577
Van der Pol	64	2.8586	1.0266	1.0248	1.3482	1.1240
Van der Pol	256	1.6541	0.8720	0.8740	1.0812	0.9448
Duffing	64	3.4597	4.2154	4.0951	2.9981	2.8227
Duffing	256	2.8364	4.0736	3.7741	2.7742	2.6921

Table 4: Controlled geometry-scaling sensitivity on all 24 original complete draws per law and both landmark counts M. Mean MSE is multiplied by 10^3. Raw inputs remain standardized; the min–max columns change only the fitted geometry scaler. Locality is the added matched coordinate control. These paired reused-draw comparisons are descriptive, with full dispersion and the score-only representation in Table 39.

7 Interpretation and limitations

What the combined evidence supports. The branch intervention shows how accurate reconstruction can coexist with a poor predictive code, and how explicit position or a locality-aware score can repair it. The fresh synthetic study shows that representation and finite decoder must be evaluated jointly: KAHM helps on a declared law while other controls win on others. The separate natural-source study shows that those controlled gains do not suffice to recommend its tested augmentation broadly. These results concern different procedures and populations; neither the natural failures erase the controlled effects nor the controlled effects override the natural failures. The later paired scaling check leaves the natural aggregate ranking intact under the specified unclipped min–max alternative. It strengthens that bounded comparison without extending it to every preprocessing choice or future source population. The main controlled local-coordinate comparisons retain their mean ordering, although Duffing's KAHM error changes appreciably. Robustness of a ranking within these two fixed scaler arms is a narrower statement than invariance of the representation or loss.

For evaluating another reconstruction-based feature map, the resulting questions are concrete. Does the code retain target-relevant position? Does a simpler locality or affine control recover the same benefit? Does the comparison persist with a raw-input control and an explicitly specified decoder? Does it survive complete selection costs and evaluation on reserved original sources? The present experiments demonstrate why each check can change the interpretation. They do not provide an automatic test for choosing a representation.

Limits of the benefit diagnostic. The four tested rules use a small set of descriptors and one reduced-capacity probe. In particular, the descriptors omit downstream landmark count; when the training cap is nonbinding, different capacities can receive identical descriptors. History and horizon also change together. Moreover, selecting by the sign of expected log loss ratio need not minimize expected scaled MSE: heterogeneous loss magnitudes matter even when each realized log ratio has the same sign as the corresponding raw-minus-KAHM loss difference. We therefore report the actual decision losses and costs as well as prediction error. The failed geometric increment motivates a new hypothesis and fresh reserved sources, rather than retuning on the already inspected confirmation outcomes.

Limits of data, inference, and computation. The eight natural source families were selected for diversity and feasibility, not sampled randomly from future applications. Four families per phase support bounded descriptive comparisons; related windows, settings, and seeds do not enlarge that number. Longer Hospital and Traffic regimes have only 21 and 29 initial training windows, and one NN5 regime retains seven test origins. Missing-window exclusion defines an available-window estimand. FRED-MD is a fixed retrospective snapshot whose upstream transformations and release-time revisions are not fully audited here.

The natural candidates are local univariate forecasters with two capacity settings and finite grids. The study is not a comprehensive neural or pretrained forecasting benchmark. The MLP reaches its iteration cap in 68 of 80 selected confirmation refits; all such fits and warnings remain reported. Equal landmark counts do not imply equal effective capacities, and the two settings jointly vary sample and model-size caps. CPU timings are implementation-specific, include material bookkeeping, and expose imports/launch separately. Process high-water RSS includes interpreter and input storage and cannot resolve incremental memory use. Model bytes, training memory, energy, and financial cost are different quantities.

The controlled Student intervals rely on their stated paired-difference assumptions, and the two contrast families retain separate coverage claims. The branch reflection and saved-bank conditioning checks are descriptive; they supply neither a population Lipschitz bound nor a distribution-free KAHM performance certificate. Replaying fixed seeds and reconstructing saved losses verifies computation without producing new source evidence.

The scaling analysis compares two fixed preprocessing choices on those same observations. It does not test clipping, all alternative scalers, local-only natural models, or a new scaler-selection rule. Near-equal natural aggregate errors do not prove statistical equivalence or scale invariance. Conversely, an observed controlled change describes the complete refitted pipeline: partitioning, truncated directions, reference-energy regularization, gates, and finite decoder geometry can all change together. Further component attribution or a newly chosen procedure would require another specified study and fresh reserved evidence for a new transfer claim.

8 Conclusion

KAHM geometry can support useful predictive features under specified target and decoder conditions. The controlled experiments locate a positional-information bottleneck, show that simpler controls can repair it, and establish conditional finite-decoder gains on Van der Pol. A separate natural-source evaluation finds higher aggregate error and measured warm-pipeline cost for the tested KAHM augmentation than for its raw-history comparator, and no useful geometric increment in the frozen benefit diagnostic. A paired post-hoc min–max check does not reverse that natural-source ranking or the main controlled mean rankings, while exposing measurable preprocessing sensitivity. The defensible practical lesson is to evaluate retained information, structural controls, the full fitting procedure, and source transfer together. A general rule for when KAHM will be worth its cost remains unestablished.

Reproducibility and research transparency

The anonymous supplement contains the unchanged original scientific dependencies, compact controlled-study records, natural-source snapshots, prepared arrays, saved per-origin forecasts, frozen diagnostics, and measured cost records. Its manifest distinguishes the post-study review export from the earlier execution freezes. The two studies can be audited separately, and their reported assets can be regenerated without selecting new models. The original natural forecast arrays reconstruct all 2,496 task–method losses. The separate scaling extension adds 1,664 natural method runs and 1,440 controlled head fits, with an execution freeze published before those fits. It retains every new natural forecast and controlled episode-loss array, plus a pre-export audit of full controlled predictions and their hashes. Replaying compact controlled losses is distinguished from recreating the omitted full prediction arrays. These computational checks establish consistency of the supplied evidence; they do not create additional independent replications or replace scientific review. The related unpublished certificate manuscript and its supplement are included anonymously to make the scope distinction inspectable.

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A Supporting Mathematical Statements

The statements in this appendix make the interpretation of the experiments precise. They are elementary supporting results; we do not claim a new KAHM approximation theorem. All conditional expectations below are understood up to almost-sure equality. For scalar square-integrable Y, write $m(X) = \mathbb{E}[Y \mid X]$ and $R_X^{\mathrm{Bayes}} = \mathbb{E}[(Y - m(X))^2]$. When a representation has been learned from independent training data, the statements apply conditionally on that fitted representation.

Proposition 1 (Information retained by a representation)**.** *For a measurable representation $U = \Gamma(X)$,*

$$R_U^{\mathrm{Bayes}} - R_X^{\mathrm{Bayes}} = \mathbb{E}[(m(X) - \mathbb{E}[m(X) \mid U])^2].$$

For any square-integrable prediction $f(U)$,

$$R(f \circ \Gamma) - R_X^{\mathrm{Bayes}} = \|m - f \circ \Gamma\|_{L^2}^2.$$

Proof. Write $Y - f(U) = (Y - m(X)) + (m(X) - f(U))$. The cross term has expectation zero after conditioning on X. Minimizing the second term over U-measurable predictions gives the conditional expectation $\mathbb{E}[m(X) \mid U]$. □

Proposition 2 (Restoring the outer score normalizer)**.** *Let $T_1, \ldots, T_C \in [0, 1]$, $\omega > 0$, and $\tau > 0$. Define*

$$\ell_c = \omega \log(1 - T_c + \tau), \qquad z = \log \sum_{j=1}^{C} \exp(\ell_j), \qquad \phi_c = \exp(\ell_c - z).$$

Then T and (ϕ, z) generate the same information in exact arithmetic, and

$$T_c = 1 + \tau - \exp\{(\log \phi_c + z)/\omega\}.$$

For multiple separately normalized blocks, one z is required per block.

Proof. Strict positivity of $1 - T_c + \tau$ implies $\phi_c > 0$. Hence $\log \phi_c + z = \ell_c$, which gives the inverse. Both directions are measurable. Applying this argument to each block gives the multiple-block statement. □

If scores are clipped before normalization, the inverse restores the clipped scores. It does not restore discarded reconstruction vectors or guarantee equal finite-head performance. In particular, the inverse is nonlinear, so affine function classes on T and on (ϕ, z) need not coincide. This proposition concerns the outer association normalizer, not the signed denominator in the KAHM reconstruction.

Proposition 3 (A sufficient condition for an injective local code)**.** *Fix a fitted bank with bases P_c and centers μ_c. Suppose its gates are measurable and strictly positive, and let A be the matrix obtained by vertically stacking the matrices $P_c^\top$. If A has full column rank d, then*

$$z \longmapsto \left[\phi(z), \{\phi_c(z) P_c^\top (z - \mu_c)\}_c\right]$$

is injective. In particular, this code and z generate the same information, and their Bayes risks agree for every square-integrable target.

Proof. The code contains both $\phi_c(z) > 0$ and its product with $P_c^\top(z - \mu_c)$. Division and addition of the fixed vector $P_c^\top \mu_c$ recover $P_c^\top z$ for every c. Stacking these vectors gives Az. Full column rank gives the measurable inverse $z = (A^\top A)^{-1} A^\top (Az)$. Conversely, the code is a measurable function of z, so the generated information and Bayes risks agree. □

The gates in Equation (2) are strictly positive in exact arithmetic because $\tau > 0$. The rank assumption is a sufficient condition, not a necessary one: when A is rank deficient, the gates themselves may retain additional information. It must be checked for each fitted bank. Very small gates and an ill-conditioned A can make the displayed inverse numerically unstable, and its existence does not make it accessible to every finite decoder or fitting procedure.

Proposition 4 (A score symmetry and its prediction consequence)**.** *Let Q be orthogonal on a preprocessed input space. Suppose that each group's reconstruction obeys $r_c(Qx) = Qr_c(x)$ and that Q preserves group identities. The radial–angular score in Section 3, its block affinities, and the block log normalizer are invariant under Q.*

For the branch law in Equation (5)*, let $Q(x_1, x_2) = (-x_1, x_2)$. Any representation $U = \Gamma(X)$ satisfying $\Gamma(QX) = \Gamma(X)$ almost surely has*

$$R_U^{\text{Bayes}} = \frac{1}{300} + \lambda + (1-\lambda)\mathbb{E}[(S - \mathbb{E}[S \mid U])^2].$$

Consequently $R_U^{\text{Bayes}} \geq 1/300 + \lambda$, with equality when U determines S.

Proof. Orthogonality preserves norms and inner products, including the zero-norm convention for the angle. Equivariance therefore preserves each radial and angular discrepancy, hence its combined score. Block normalization and its log normalizer are deterministic functions of those scores.

For the branch law, replacing T by $-T$ leaves the joint distribution of (S, T) unchanged and leaves U unchanged. Thus $\mathbb{E}[T \mid U] = 0$. Also S is even under this reflection, whereas T is odd. It follows that $\mathbb{E}[(S - \mathbb{E}[S \mid U])T] = 0$. Expanding the square in Proposition 1, using $\mathbb{E}[T^2] = 1/3$ and $\text{Var}(\epsilon) = 1/300$, yields the displayed identity. □

The first assertion also holds when Q permutes reconstructors within a group whose score is their minimum. Permuting distinct groups, however, can permute the affinity coordinates rather than leaving the representation unchanged. Nor does a raw-input reflection necessarily commute with a training-fitted scaler. Finite learned partitions and reference sets need not satisfy the proposition's assumptions. The measured near-invariance of one learned code is therefore a diagnostic, not a proof that its Bayes risk equals the exact-symmetry lower bound. Tiny nonzero differences could in principle be used by an unrestricted decoder.

Proposition 5 (Approximate symmetry under a decoder budget)**.** *Let Q be a measurable, distribution-preserving involution on the input space: $Q^2X = X$ almost surely and QX has the law of X. Let d_U be a measurable pseudometric on the representation space and g be κ-Lipschitz for this pseudometric. Assume $m, g \circ \Gamma \in L^2$ and*

$$\delta_\Gamma^2 = \mathbb{E}[d_U(\Gamma(X), \Gamma(QX))^2] < \infty.$$

With $m_o = (m - m \circ Q)/2$,

$$R(g \circ \Gamma) - R_X^{\text{Bayes}} \geq (\|m_o\|_{L^2} - \kappa\delta_\Gamma/2)_+^2.$$

For the branch law, $\|m_o\|_{L^2} = \sqrt{\lambda}$.

Proof. The operator $Vh = h \circ Q$ is an isometric involution on $L^2(P_X)$. The even and odd components $h_e = (h + Vh)/2$ and $h_o = (h - Vh)/2$ are orthogonal. In particular, with $f = g \circ \Gamma$,

$$\|m - f\|_{L^2}^2 = \|m_e - f_e\|_{L^2}^2 + \|m_o - f_o\|_{L^2}^2 \geq \|m_o - f_o\|_{L^2}^2.$$

The Lipschitz condition gives $|f_o(X)| \leq \kappa d_U(\Gamma(X), \Gamma(QX))/2$, so $\|f_o\|_{L^2} \leq \kappa\delta_\Gamma/2$. Apply the reverse triangle inequality and Proposition 1. For the branch law, $m_o(X) = \sqrt{3\lambda}\,T$ and $\mathbb{E}[T^2] = 1/3$. □

An empirical reflection distance is not automatically a population upper bound for δ_Γ. A valid numerical lower bound obtained from this proposition would need uncertainty control in that direction and a verified decoder Lipschitz constant for the same metric. Neither is asserted for the experiment's unconstrained diagnostic heads.

Proposition 6 (Reconstruction through the predictor's representation)**.** *Suppose a version of m is L-Lipschitz on a domain containing X and $h(U)$ almost surely, and $\mathbb{E}\|X - h(U)\|_2^2 \leq \varepsilon_{\text{rec}}^2$. Then*

$$R_U^{\text{Bayes}} - R_X^{\text{Bayes}} \leq L^2\varepsilon_{\text{rec}}^2.$$

If h is additionally L_h-Lipschitz from the declared representation metric and $\kappa \geq LL_h$, the same upper bound applies to the infimum excess risk over κ-Lipschitz decoders:

$$\inf_{\mathrm{Lip}(g)\leq\kappa} \{R(g \circ \Gamma) - R_X^{\mathrm{Bayes}}\} \leq L^2 \varepsilon_{\mathrm{rec}}^2.$$

If a particular decoder f instead satisfies $\|m \circ h - f\|_{L^2(P_U)} \leq \delta_{\mathrm{head}}$, then

$$R(f \circ \Gamma) - R_X^{\mathrm{Bayes}} \leq (L\varepsilon_{\mathrm{rec}} + \delta_{\mathrm{head}})^2.$$

Proof. Conditional expectation is the L^2 projection onto the U-measurable functions. The admissible prediction $m(h(U))$ gives

$$\mathbb{E}[(m(X) - \mathbb{E}[m(X) \mid U])^2] \leq \mathbb{E}[(m(X) - m(h(U)))^2] \leq L^2 \mathbb{E}\|X - h(U)\|_2^2.$$

For the constrained statement, $m \circ h$ has Lipschitz constant at most LL_h. For a particular f, the triangle inequality in L^2 gives $\|m - f \circ \Gamma\|_{L^2} \leq L\varepsilon_{\mathrm{rec}} + \delta_{\mathrm{head}}$. Use Proposition 1 in each case. □

The essential assumption is the existence of h as a function of the retained representation. A KAHM reconstruction evaluated on the raw query is not such a function merely because its residual is small. The domain assumption also matters: a conditional mean initially defined only on the support of X must have the specified Lipschitz version at $h(U)$.

Proposition 7 (Common scaling of an inner kernel row)**.** *Let k be a finite kernel row, let B be the fixed fitted matrix, and suppose $kB\mathbf{1} \neq 0$. For every real $a > 0$,*

$$\frac{akB}{akB\mathbf{1}} = \frac{kB}{kB\mathbf{1}}.$$

Thus evaluating a Gaussian query row by $\exp(\ell_j - \max_i \ell_i)$ preserves its normalized signed weights in exact arithmetic whenever the original denominator is nonzero.

Proof. The common factor cancels from numerator and denominator. For finite log-kernel values, the centered exponential row is the original row times $a = \exp(-\max_i \ell_i) > 0$ in real arithmetic. □

This statement neither rescales the training kernel nor permits replacing B. It does not address a genuinely zero signed denominator or prevent cancellation. The purpose of the implemented change is to avoid evaluating an entire query row as floating-point zero before normalization.

B Generators, Shared Components, and Decoders

B.1 Exact forecasting laws and target construction

The nonlinear AR generator starts with independent $u_0, u_1 \sim \mathrm{Unif}[-0.8, 0.8]$ and uses

$$u_{t+1} = \tanh\{1.25u_t - 0.65u_{t-1} + 0.25\sin(3u_t) + \eta_{t+1}\}, \qquad \eta_t \overset{\mathrm{iid}}{\sim} \mathrm{Unif}[-0.1, 0.1].$$

The first 80 stored values are discarded; the next 72 are the episode. Each episode uses an independent initial state and innovation sequence.

The Van der Pol and Duffing systems respectively use

$$\dot{x} = v, \qquad \dot{v} = (1 - x^2)v - x,$$
$$\dot{x} = v, \qquad \dot{v} = -0.2v + x - x^3 + 0.3\cos(1.2t + \theta).$$

Initial positions and velocities are independent $\mathrm{Unif}[-2, 2]$ draws for each episode, and $\theta \sim \mathrm{Unif}[0, 2\pi]$ is independently sampled for each episode. The phase draw is unused for Van der Pol but is retained in the

common generator's random-stream order. Both systems are integrated using float64 classical fourth-order Runge–Kutta with step 0.025, burn-in time 10, and observation interval 0.1. At the 72 observation times, the sensor is

$$u_t = 0.95\tanh(x_t) + \eta_t, \qquad \eta_t \overset{\text{iid}}{\sim} \text{Unif}[-0.02, 0.02].$$

Sensor noise is sampled before numerical integration in the frozen generator. These define numerical data laws. We do not treat the resulting samples as exact solutions of the continuous equations or assume stationarity after burn-in.

For all three laws, zero-based forecast origins are $11, 15, \ldots, 59$. At origin t, the input is $(u_{t-11}, \ldots, u_t)$ and the three targets are $(u_{t+1}, u_{t+4}, u_{t+12})$. There are 13 origins per episode. Training, validation, and test contain respectively 128, 128, and 512 independently generated episodes. They therefore contain 1,664, 1,664, and 6,656 rows, but rows within an episode are dependent.

The branch law is Equation (5). Training has 1,024 points for the primary study, with the first 256 points giving the secondary nested sample-size condition. Validation and test have 512 and 2,048 independent points. At each repetition, the same latent draws of S, T, ϵ generate both curvatures; within each curvature and training size, one fitted bank serves all three target mixtures. Latent labels never train the partitions. Since $T = X_1$ and $S = 2\{X_2 - c\cos(\pi X_1)\}$, the raw inputs determine the signal. Its variance is $(1-\lambda) + 3\lambda/3 = 1$, and the independent noise variance is $1/300$. The support bound $|Y| \le \sqrt{1-\lambda} + \sqrt{3\lambda} + 0.1 \le 2.1$ justifies the common clipping range.

The master random seed is 20261030. Separate phase identifiers and child streams determine pilot, first confirmation, natural fitting, and fresh capacity inputs and fits. The capacity phase identifier is 4. The archive contains the executable stream rules and the realized identifiers. Reusing a replicate label across phases does not reuse its observations. Common randomness within a repetition supports pairing; it does not create additional independent repetitions.

B.2 Training the common bank and defining the locality control

The input mean and population-standard-deviation convention ($1/n$ variance divisor) are fitted on all training rows. A standard deviation at most 10^{-12} is replaced by one. At most 1,600 training rows, sampled without replacement, are used for clustering and reference selection. Each of the three independent k-means block fits uses five initializations and at most 150 iterations. Its cluster counts are 2, 4, and 8. At most 24 reference rows per cluster are sampled without replacement and converted to float32 before the unchanged upstream autoencoder is called. The effective requested rank is $\min(2, d-1, n_c-1)$, subject to upstream numerical rank determination. The common reference mean and the resulting upstream principal basis define the affine control.

For completeness, the pinned local KAHM fit can be written as follows. Let $H \in \mathbb{R}^{d\times n_c}$ contain the float32 reference columns, and set $q_i = P_c^\top H_i$. The basis is obtained from the centered reference covariance with divisor $n_c - 1$, retaining eigenvalues above machine epsilon. The requested rank is reduced if a retained projected coordinate has range below 10^{-3}, until that condition is removed or rank one is reached. With effective rank r_c and unbiased projected covariance Σ_c, the training kernel is

$$K_{ij} = \exp\left\{-\frac{(q_i - q_j)^\top \Sigma_c^{-1}(q_i - q_j)}{2r_c}\right\}.$$

Writing $\nu = \|H\|_F^2/(dn_c)$, initialize $\rho_0 = 2.5\nu$ and iterate

$$B_j = (K + \rho_j I)^{-1}, \qquad \rho_{j+1} = 2\nu + \frac{\|H - H(KB_j)^\top\|_F^2}{dn_c}.$$

The routine stops when $|\rho_{j+1} - \rho_j| \le 10^{-3}$ or after 100 iterations, returning the last computed B_j. This regularizer is determined from reference inputs alone, independently of the supervised target. Query kernels use the same projected covariance metric and $1/(2r_c)$ Gaussian scale. A cluster with fewer than two usable references or no positive empirical-variance direction has the recorded center reconstruction fallback; other numerical failures are not silently replaced by another method.

The two reconstruction methods have the same input basis but not identical functional capacity. KAHM uses its nonlinear query weights to combine full-dimensional reference vectors; it need not remain inside the truncated affine PCA plane. A shared basis does not make its reconstruction an affine projection. The experiment isolates the reconstruction operator within the specified bank, not all possible advantages of independently optimized complete systems.

The alternative affine locality score is essential to interpreting the straight-branch comparison. In standardized input space, let

$$s_c = \max\left\{\frac{1}{n_c}\sum_{i\in c}\|z_i-\mu_c\|_2^2, 10^{-6}\right\},$$
$$v_c = \max\left\{\frac{1}{n_c}\sum_{i\in c}\|(I-P_cP_c^\top)(z_i-\mu_c)\|_2^2, 0.05s_c, 10^{-6}\right\}.$$

It uses

$$b_c(z) = \frac{\|z-r_c^{\mathrm{A}}(z)\|_2^2}{v_c} + 0.1\frac{\|z-\mu_c\|_2^2}{s_c}, \qquad \psi_c(z) = \frac{\exp(-b_c(z))}{\sum_{j \text{ in block}}\exp(-b_j(z))}.$$

This retains a center-distance term that distinguishes positions along an otherwise exactly reconstructed line. It changes the scoring rule and is a secondary control, not part of the primary reconstruction substitution. A radial-basis locality control similarly normalizes $\exp\{-\|z-\mu_c\|_2^2/(2\,\mathrm{median}_j\, s_j)\}$ within a block.

There are 14 stored affinity coordinates and three block-sum constraints, giving an affine-dimension upper bound of 11 for the concatenated affinities. Appending the three log normalizers gives 17 stored coordinates. Local augmentation gives 28 coordinates for the two-dimensional branches and 42 for the 12-dimensional forecasting histories. The added coordinates are $\phi_c P_c^\top(z-\mu_c)$, computed from the original query in the shared basis; full reconstruction vectors are not passed to these heads.

For reconstruction diagnostics, each query's squared reconstruction distance is divided by input dimension. The best-model diagnostic takes the minimum across local models in each block; the weighted diagnostic takes the sum with that reconstruction's own affinity weights. Both then average across queries, blocks, and repetitions. Thus the weighted summary is an affinity-weighted average of individual errors, not the reconstruction error of the affinity-weighted vector. These descriptive summaries are not additional primary forecast contrasts.

B.3 Ridge and finite nonlinear heads

Each input feature column is standardized using training data alone. Standard deviations below 10^{-8} are replaced by one. The ridge fit minimizes mean squared training error plus $\alpha\|W\|_F^2$, with an unpenalized intercept. The fixed penalty grid is

$$\alpha \in \{10^{-8}, 10^{-6}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}.$$

Validation chooses the penalty. Forecast horizons share that choice; the three branch targets choose separately.

The nonlinear head first selects M training rows without replacement as landmarks in the standardized input-feature space. For a fixed budget, the common seed selects the same row indices across the representations, though their feature values differ. Let b^2 be the median strictly positive squared landmark distance, excluding distances at most 10^{-14}; if no such distance exists, use $b^2 = 1$. For bandwidth multiplier $a \in \{0.25, 1, 4\}$, the squared bandwidth is $\max(b^2a^2, 10^{-12})$. Form the Gaussian landmark matrix K_{MM}, symmetrize it, and retain eigenvalues exceeding $\max(10^{-10}\lambda_{\max}, 10^{-12})$. If $K_{MM} = V\Lambda V^\top$ on the retained subspace, the nonlinear coordinates are $K_{xM}V\Lambda^{-1/2}$. Concatenate these coordinates with the standardized linear input features; standardize the full resulting basis on training data and fit the same ridge objective. Validation selects a bandwidth–penalty pair using the fixed nested iteration order, so ties are deterministic. The first stage uses $M = 64$; the fresh follow-up uses both 64 and 256.

Because basis columns are standardized and assigned an ordinary Euclidean ridge penalty, this is a finite nonlinear diagnostic with a linear component. It is not an exact kernel-ridge estimator with its standard RKHS

norm. Effective dimension can be smaller than input dimension plus M after the eigenvalue cutoff. Increasing M can change the sampled landmark set, median-distance bandwidth, retained rank, and validation-selected hyperparameters. It does not isolate only a dimension parameter while holding the realized basis fixed.

Predictions are clipped to the known target range for both validation and test evaluation, but the ridge objective itself is the un-clipped training least-squares objective. The selected coefficients remain fitted on training data; validation observations are not used in a final combined fit. The secondary capacity selector chooses the lowest validation MSE among the eight complete representation–budget pipelines. Its fixed tie order is ascending budget followed by the configured feature order. The test split chooses neither hyperparameters nor the complete pipeline.

B.4 Airfoil grouping and the scope of its repetitions

The Airfoil snapshot contains 1,503 rows with five covariates and sound pressure as the target. Exact joint equality of the four non-frequency covariates defines 106 operating-setting groups. A fixed permutation with seed 2026103007 allocates 64, 21, and 21 groups to training, validation, and test, giving 905, 295, and 303 rows. All frequency rows within a setting stay together. The training objective weights rows equally; validation and test average within a group before weighting groups equally. The outcomes are reported in squared decibels.

The 24 seeds repeat fitting on this same observed cohort and split. They characterize fitting variability conditional on those data. No primary synthetic-experiment interval is applied to Airfoil, and the 24 fits are not treated as 24 independent draws of natural tasks, test cohorts, or physical operating-condition populations. A population claim would require a justified sampling model or independent cohorts.

C Estimands, Multiplicity, and Descriptive Diagnostics

C.1 The unit of paired inference

For forecast method A and repetition r, the reported loss is

$$\widehat{L}_{r,A} = \frac{1}{512}\sum_{j=1}^{512}\frac{1}{13\cdot 3}\sum_{o=1}^{13}\sum_{h\in\{1,4,12\}}(\widehat{y}_{r,A,j,o,h} - y_{r,j,o,h})^2.$$

For branches it is the mean noisy-target squared error over 2,048 test points. The conditional-mean target loss is a separate secondary diagnostic. Each method and target at a given repetition shares the declared test draws, yielding a paired contrast. The independent units for the primary intervals are the 24 complete train–validation–test repetitions. Overlapping windows, horizons, reference groups, candidate fits, and three target mixtures are not extra replication units.

For each primary paired contrast D_r, let $\overline{D}$ be its sample mean and s_D its unbiased sample standard deviation over 24 repetitions. For family size F, the interval is

$$\overline{D} \;\pm\; t_{23,\,1-0.05/(2F)}\,\frac{s_D}{\sqrt{24}}.$$

The critical values are approximately 3.545808 for $F = 29$ and 3.180663 for $F = 12$. Under independent normal paired differences, each marginal Student interval has its exact nominal coverage, and the Bonferroni union bound gives family coverage at least 95%. Under other laws, the Student coverage is approximate. Twenty-four repetitions do not guarantee an adequate normal approximation. The results are not distribution-free certificates. Dependence between contrasts is allowed by Bonferroni; independence of windows is never used.

The synthetic estimand includes training and validation randomness, as well as finite test-sample variation, under each declared numerical generator. It is not an average over an independently sampled population of forecasting tasks. The sample size of 24 was fixed as a resource decision, not justified by an unreported prospective power calculation.

C.2 Primary controlled contrasts

The first stage compares geometry, normalizer retention, and local-coordinate retention within each of its nine settings:

$$G = R(\phi^{\mathrm{A}}, \mathrm{ridge}) - R(\phi^{\mathrm{K}}, \mathrm{ridge}),$$
$$N = R(\phi^{\mathrm{K}}, \mathrm{ridge}) - R([\phi^{\mathrm{K}}, \log Z^{\mathrm{K}}], \mathrm{ridge}),$$
$$L = R(\phi^{\mathrm{K}}, \mathrm{ridge}) - R(\Gamma^{\mathrm{K}}_{\mathrm{local}}, \mathrm{ridge}).$$

Here R denotes the test loss of the indicated complete fitted procedure. Positive values favor the second procedure in each difference. The 27 cellwise contrasts and $G(c,0) - G(c,1)$ at the two curvatures form the 29-member family. On straight branches, the position-target geometry advantage is 0.978026 with simultaneous interval $[0.96255, 0.99350]$. The target interaction is -0.978039 with interval $[-0.99351, -0.96257]$: the advantage is larger when position matters. This direction is not evidence that branch identity favors KAHM.

The normalizer inverse is checked numerically to about 2.8×10^{-16}. Nonetheless, the first-stage Van der Pol full-score RBF MSE is 0.00158, versus 0.01273 for the information-equivalent $[\phi, \log Z]$ coordinates. These secondary fitted-head comparisons illustrate accessibility and regularization differences; they do not estimate the representations' Bayes risks.

C.3 Comparison families and practical margins

The first-stage family contains the 27 G, N, L contrasts over nine settings and the two interactions $I_c = G(c, \lambda = 0) - G(c, \lambda = 1)$. A positive interaction would favor a larger reconstruction advantage for the branch-identity target. Negative I_c is the opposite direction of this hypothesis, not a literal statement that one fitted method is uniformly harmful. The cell margin is 5% of the mean raw-ridge MSE from two pilot repetitions. Each interaction margin is the mean of its two endpoint cell margins. All pilot outcomes are excluded from confirmation estimates. The margins are recorded design thresholds, not externally validated application tolerances; their scale estimate has only two pilot observations.

The fresh capacity family contains 12 differences: raw minus KAHM local and affine local minus KAHM local, for three systems and two budgets. Each law's margin is 5% of its first-stage mean raw-RBF MSE, using the complete 24 first-stage confirmation repetitions. The margins, candidate list, and 24 fresh replication streams were frozen before phase-4 data were generated. Conditional on the completed first stage, they are fixed for this follow-up. The follow-up is explicitly data-motivated; it is not retroactively part of the initial 29-comparison plan. Its separate 95% family does not imply joint 95% coverage with the first family.

We call a difference a practical benefit at the declared threshold only when its simultaneous lower endpoint exceeds its fixed margin. An interval strictly above zero but not above the margin establishes a positive difference under the interval assumptions without resolving that practical threshold. A negative interval indicates an adverse difference for the candidate. An interval containing zero leaves the direction unresolved. An interval entirely within the fixed band $[-\delta, \delta]$ can support that band; failure to exclude zero alone cannot. Percent changes in the paper are descriptive ratios of means, without an asserted ratio confidence interval.

The capacity family has no primary interaction testing whether the advantage changes with landmark budget. A positive interval at one budget and an unresolved interval at another would not prove a budget interaction. All declared cells, including adverse and unresolved results, are retained. There were no omitted failed prescribed repetitions in the completed archives.

C.4 Reflection, reconstruction, and numerical checks

Reflection is applied to the raw branch input before the frozen scaler and geometry. For a p-coordinate feature vector, the stored-coordinate diagnostic is $\{\mathbb{E}_n \|\Gamma(X) - \Gamma(QX)\|_2^2 / p\}^{1/2}$. The standardized diagnostic first divides each coordinate by its training-only standard deviation, replacing values at most 10^{-12} by one. This threshold differs from the fitted heads' 10^{-8} rule. Thus the reported per-coordinate RMS, its standardization, and its finite-sample approximation are not silently identified with the population metric or a known Lipschitz

budget in Proposition 5. Values in Table 12 are averages of per-repetition RMS diagnostics, not a single pooled RMS.

The post-hoc numerical check retains all 24 phase-4 Van der Pol banks and all train/validation/test observations. It changes only query-row Gaussian evaluation as in Proposition 7, before multiplication by the unchanged fitted B. The training kernel, regularizer, principal basis, reference values, covariance metric, partition, and scaler are unchanged. Heads are fitted on the stabilized training features and retuned on the same validation split. The 96 outcomes cover two encodings, two budgets, and 24 repetitions. They are descriptive reuse of phase-4 data; no third confirmatory family or new confidence claim is attached.

The independently checked test count of 99,762 zero normalizers refers to query–local-model pairs among 2,236,416 evaluations, not distinct test observations. Of those pairs, 99,687 already had all Gaussian entries equal to floating-point zero; the remaining 75 zeros arose in later arithmetic. The stabilized count was zero. Local-feature losses were effectively unchanged, while full-score losses and some predictions changed. This addresses that particular underflow pathway as an explanation of the local-feature benefit; it is not a proof of numerical robustness of every KAHM encoding or of signed normalization in general.

C.5 Retrospective influence and coordinate-recovery diagnostics

These descriptive checks were added after the original study. They reuse the declared paired repetitions and do not change the primary estimates, margins, or comparison families. For each of the 41 primary contrasts, we compute bias-corrected sample skewness and delete each repetition once. Each deletion recomputes the Student interval with 23 remaining repetitions, 22 degrees of freedom, and the original family size (29 or 12). The range of deletion endpoints is an influence diagnostic, not a confidence interval or an additional simultaneous-coverage claim. Table 5 shows the four Van der Pol comparisons; the complete descriptive results accompany the executable artifact. All four Van der Pol comparisons retain lower endpoints above the fixed practical margin after any one-repetition deletion. This does not establish a distribution-free mean guarantee or remove the normal-approximation qualification of the primary analysis.

Table 5: Retrospective Van der Pol influence diagnostics. Lower endpoints are multiplied by 10^3; the fixed margin on this scale is 0.148907. Skewness is the bias-corrected Fisher–Pearson sample coefficient. The minimum deletion lower endpoint is the smallest over all 24 one-repetition deletions. The final column counts individual paired differences exceeding the margin; it is not a mean-effect confidence statement.

Control	M	Skewness	Original lower	Min. deletion lower	$D_r > \delta$
Raw	64	-0.07	1.7369	1.7278	24/24
Affine local	64	2.69	0.2755	0.2685	24/24
Raw	256	0.01	0.7550	0.7524	24/24
Affine local	256	0.33	0.1810	0.1786	23/24

We also inspect the 120 first-stage primary banks and 72 capacity banks from a completed computational reproduction using the published seeds. No bank or head is refitted for these diagnostics. Replayed test inputs, targets, and groups are checked against their recorded hashes before the original feature map is evaluated. Let A vertically stack the $P_c^\top$ matrices. Proposition 3 shows that positive gates and full column rank of A suffice to recover the scaled input from the gated local coordinates and gates: division by ϕ_c recovers $P_c^\top(z - \mu_c)$, after which one solves for z. A deficient stack does not establish information loss in the full code, because the gates can carry additional information. Numerical ranks use the threshold $\sigma_1 \max\{\mathrm{rows}(A), d\}\epsilon_{64}$; they are finite-precision diagnostics, not exact rank proofs. Small singular values and small gates can make coordinate recovery sensitive to perturbations. Test-sample gate minima are not population lower bounds, and these computations do not guarantee a well-conditioned inverse or easy decoding by any particular fitted head.

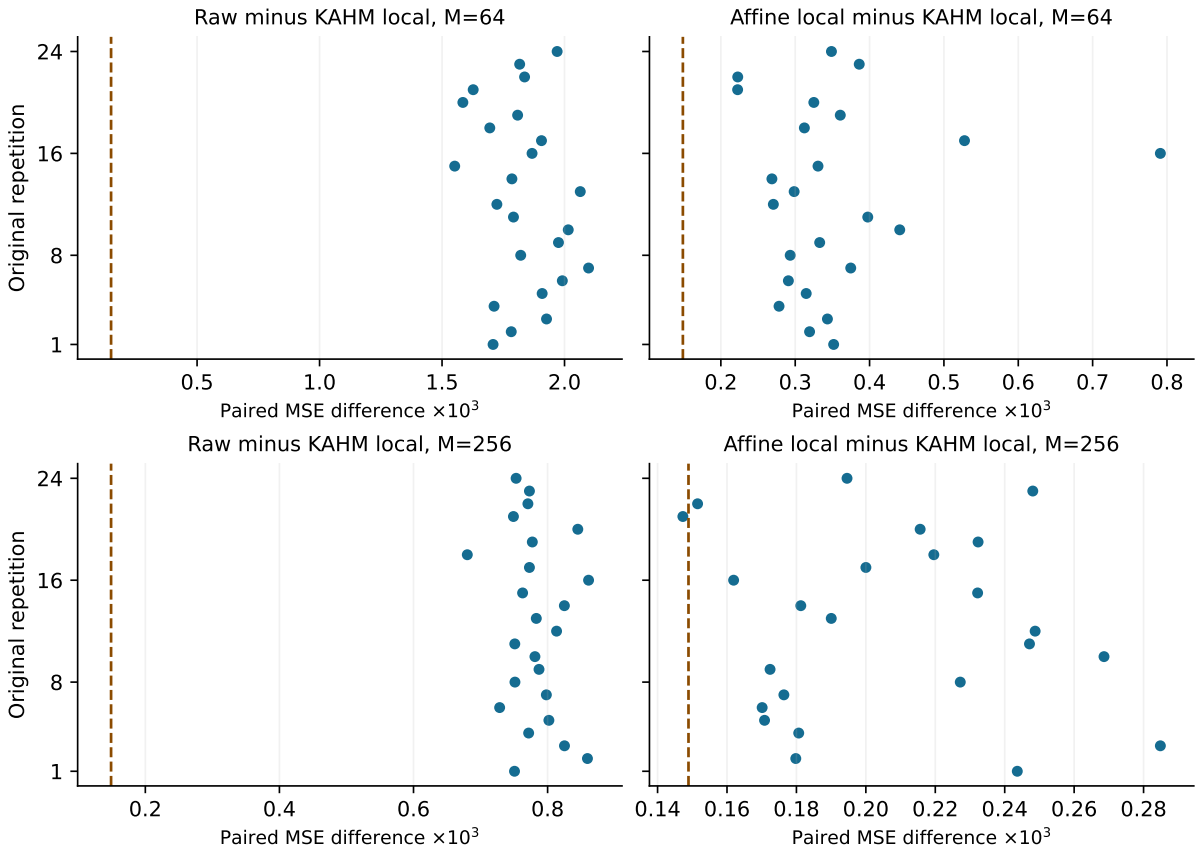


Figure 2: The 24 original paired differences for each Van der Pol capacity comparison, in units of 10^{-3} MSE. The dashed line is the frozen practical margin. No resampling, refitting, or new primary analysis is performed.

Table 6: Retrospective diagnostics over 24 stored banks per row. Rank ranges concern the stacked local bases in input dimension d. The smallest singular value and gate entries are minima over the inspected banks (and, for gates, their original test observations); κ_2 is the largest stack condition number. A dash means that not every stack in that row has full numerical column rank. KAHM and affine representations share the basis stack. Computational reproduction supplies no new independent statistical replications.

Phase / setting	Rank / d	Min. σ_d	Max. κ_2	Min. ϕ^{K}	Min. ϕ^{A}
First: Straight branch	1/2	0	—	3.36×10^{-25}	2.56×10^{-20}
First: Curved branch	2/2	2.19×10^{0}	1.39×10^{0}	3.55×10^{-28}	8.97×10^{-29}
First: Nonlinear AR	12/12	6.20×10^{-2}	4.90×10^{1}	2.21×10^{-35}	2.36×10^{-34}
First: Van der Pol	12/12	1.60×10^{-2}	2.20×10^{2}	2.84×10^{-45}	9.43×10^{-20}
First: Duffing	12/12	1.33×10^{-2}	2.80×10^{2}	3.92×10^{-43}	3.11×10^{-7}
Capacity: Nonlinear AR	12/12	7.31×10^{-2}	4.19×10^{1}	9.97×10^{-37}	7.75×10^{-37}
Capacity: Van der Pol	12/12	1.60×10^{-2}	2.19×10^{2}	6.21×10^{-44}	1.56×10^{-16}
Capacity: Duffing	12/12	1.32×10^{-2}	2.84×10^{2}	5.38×10^{-43}	1.86×10^{-8}

D Complete controlled results and diagnostics

Table 7: The six primary branch settings at 1,024 training points: mean ridge test MSE over 24 repetitions, with the primary KAHM advantage G and its interval from the 29-comparison family. The Bayes MSE is 1/300. The affine-locality scoring control reuses the shared references and is secondary.

c	λ	Affine ϕ	KAHM ϕ	Affine locality	G [interval]
0	0	0.003334	0.003347	0.003344	−0.000013 [−0.000031, 0.000005]
0	0.5	0.503448	0.013995	0.003991	0.489453 [0.481962, 0.496944]
0	1	1.002694	0.024668	0.004641	0.978026 [0.962552, 0.993500]
1	0	0.282887	0.281126	0.314394	0.001761 [−0.071260, 0.074781]
1	0.5	0.160813	0.165207	0.167971	−0.004395 [−0.043672, 0.034882]
1	1	0.062366	0.051042	0.045555	0.011324 [0.000882, 0.021767]

Table 8: All nine first-stage primary settings: mean test MSE over 24 repetitions for ridge decoders. The local-coordinate improvement L and its interval belong to the 29-comparison family. Retaining local position substantially improves KAHM prediction in eight settings; KAHM local coordinates do not dominate affine local coordinates.

Setting	KAHM ϕ	KAHM local	Affine local	Raw	L [interval]
$c = 0, \lambda = 0$	0.003347	0.003378	0.003337	0.003338	−0.000031 [−0.000043, −0.000020]
$c = 0, \lambda = 0.5$	0.013995	0.003379	0.003333	0.003334	0.010616 [0.010033, 0.011200]
$c = 0, \lambda = 1$	0.024668	0.003380	0.003333	0.003330	0.021287 [0.020199, 0.022376]
$c = 1, \lambda = 0$	0.281126	0.017982	0.048916	0.671776	0.263144 [0.217187, 0.309101]
$c = 1, \lambda = 0.5$	0.165207	0.010497	0.026756	0.337620	0.154710 [0.122867, 0.186552]
$c = 1, \lambda = 1$	0.051042	0.004188	0.004656	0.003329	0.046854 [0.040400, 0.053307]
Nonlinear AR	0.005300	0.004867	0.004867	0.005053	0.000433 [0.000368, 0.000498]
Van der Pol	0.111108	0.014475	0.011582	0.041340	0.096633 [0.092436, 0.100830]
Duffing	0.053729	0.007560	0.004448	0.006764	0.046168 [0.041729, 0.050608]

Table 9: Post-hoc Van der Pol numerical sensitivity on the same 24 second-stage banks and samples. Stable query evaluation removes observed zero denominators while leaving local-coordinate mean MSE effectively unchanged. Δ_r is the original-minus-stable MSE difference within repetition r; wins count stable predictions beating the raw and affine local controls at the same width. These comparisons are outside the two primary interval families.

Encoding	M	Original MSE	Stable MSE	Max. $\lvert\Delta_r\rvert$	Wins raw; affine
Local	64	0.001026627	0.001026627	1.01×10^{-12}	24/24; 24/24
Local	256	0.000871969	0.000871969	1.69×10^{-11}	24/24; 24/24
Scores	64	0.001571085	0.001565219	8.74×10^{-5}	24/24; 4/24
Scores	256	0.001049909	0.001045801	3.53×10^{-5}	24/24; 17/24

Tables 11 and 12 give descriptive input-geometry summaries. The two complete primary comparison families follow. Their values are generated from the archived outcomes, with each difference, standard error, interval, and frozen margin independently recomputed by the paper-asset script before it writes these tables.

Table 10: Secondary Airfoil static-regression results on one fixed group split, averaged over 24 fitting seeds. Units are dB^2. These seeds do not constitute independent natural-data replications, and the comparison is descriptive.

Representation	Ridge MSE	RBF_{64} MSE
Raw	19.4726	15.6898
Affine ϕ	34.1630	36.0476
KAHM ϕ	31.8586	34.3338
Affine local	22.7200	22.4050
KAHM local	20.9040	20.9313
KAHM scores	24.2168	21.7070

Table 11: Held-out reconstruction errors per standardized input coordinate, averaged over the three blocks and 24 full repetitions. "Best" takes the best local reconstruction separately in each block; "weighted" averages the individual squared errors with that method's affinities. The affine value is lower in all 24 repetitions of each of these five input settings, for both summaries. These are descriptive diagnostics, not the supervised primary endpoint.

Setting	KAHM best	Affine best	KAHM weighted	Affine weighted
Branch $c = 0$	3.276×10^{-3}	2.156×10^{-16}	4.006×10^{-3}	1.548×10^{-8}
Branch $c = 1$	3.583×10^{-2}	2.932×10^{-2}	3.793×10^{-2}	3.238×10^{-2}
Nonlinear AR	1.166×10^{-2}	1.084×10^{-2}	1.305×10^{-2}	1.214×10^{-2}
Van der Pol	4.353×10^{-3}	1.683×10^{-3}	4.495×10^{-3}	1.916×10^{-3}
Duffing	4.877×10^{-3}	5.613×10^{-4}	5.239×10^{-3}	1.226×10^{-3}

Table 12: Branch reflection differences at training size 1,024, averaged over 24 banks. The statistic is a per-coordinate RMS, with and without training-only coordinate scaling. The raw reflection precedes the frozen pipeline. These values neither establish exact information loss for the learned code nor certify a decoder-specific population risk bound.

c	Encoding	Stored-coordinate RMS	Standardized RMS
0	KAHM ϕ	4.894×10^{-1}	1.402×10^{0}
0	Affine ϕ	1.451×10^{-9}	7.533×10^{-9}
0	KAHM local	5.289×10^{-1}	1.418×10^{0}
0	Affine local	4.993×10^{-1}	1.236×10^{0}
1	KAHM ϕ	5.380×10^{-1}	1.427×10^{0}
1	Affine ϕ	5.321×10^{-1}	1.471×10^{0}
1	KAHM local	4.854×10^{-1}	1.442×10^{0}
1	Affine local	5.522×10^{-1}	1.473×10^{0}

Table 13: Complete representation means in the fresh capacity study, with MSE multiplied by 10^3. The full-score KAHM representation is a prespecified secondary control. These point estimates do not add hypotheses to the 12-comparison primary family.

System	M	Raw	Affine local	KAHM local	KAHM scores
Nonlinear AR	64	4.4238	4.9120	4.8795	4.7984
Nonlinear AR	256	4.6958	5.2347	5.1749	4.8256
Van der Pol	64	2.8586	1.3767	1.0266	1.5711
Van der Pol	256	1.6541	1.0780	0.8720	1.0499
Duffing	64	3.4597	3.0549	4.2154	4.7005
Duffing	256	2.8364	2.7981	4.0736	3.3808

D.1 Selected pipelines and realized dimensions

The validation-selected pipeline and the retained nonlinear basis are distinct summaries: the former describes the operational model choice, whereas the latter records the realized dimensions behind the nominal landmark budgets.

Table 14: Secondary validation-only selection among the eight complete capacity-study pipelines. Mean test MSE is multiplied by 10^3. Counts concern independent complete repetitions under each generator, not a new population of forecasting tasks.

System	Selector MSE	Validation selections
Nonlinear AR	4.4238	Raw, 64: 24/24
Van der Pol	0.8720	KAHM local, 256: 24/24
Duffing	2.7923	Affine local, 256: 23/24; Raw, 256: 1/24

Table 15: Retained RBF-plus-linear dimensions after the Nyström eigenvalue cutoff in the fresh capacity study. Means and ranges cover the 24 repetitions in each cell. The same landmark count M does not imply the same realized dimension or function class.

System	M	Representation	Mean dimension	Range
Nonlinear AR	64	Raw	74.88	61–76
Nonlinear AR	64	Affine local	106.00	106–106
Nonlinear AR	64	KAHM local	106.00	106–106
Nonlinear AR	64	KAHM scores	76.71	62–78
Nonlinear AR	256	Raw	189.04	78–268
Nonlinear AR	256	Affine local	297.62	296–298
Nonlinear AR	256	KAHM local	297.50	293–298
Nonlinear AR	256	KAHM scores	148.88	77–219
Van der Pol	64	Raw	76.00	76–76
Van der Pol	64	Affine local	106.00	106–106
Van der Pol	64	KAHM local	106.00	106–106
Van der Pol	64	KAHM scores	78.00	78–78
Van der Pol	256	Raw	268.00	268–268
Van der Pol	256	Affine local	298.00	298–298
Van der Pol	256	KAHM local	272.38	261–285
Van der Pol	256	KAHM scores	231.88	215–251
Duffing	64	Raw	76.00	76–76
Duffing	64	Affine local	106.00	106–106
Duffing	64	KAHM local	105.46	98–106
Duffing	64	KAHM scores	77.96	77–78
Duffing	256	Raw	268.00	268–268
Duffing	256	Affine local	296.92	272–298
Duffing	256	KAHM local	240.38	176–298
Duffing	256	KAHM scores	214.33	164–270

Table 16: Complete first-stage family (29 contrasts), in original MSE units. Intervals use the two-sided 95% Bonferroni paired-replicate Student procedure with 23 degrees of freedom. Under paired normality, each t interval is exact and Bonferroni family coverage is at least 95%; otherwise coverage is approximate. The two stages have separate families.

Setting / budget	Contrast	$\overline{D}$	Simultaneous interval	δ	Assessment
$c = 0, \lambda = 0$	G	-0.000013	$[-0.000031, 0.000005]$	0.000168	unresolved †
$c = 0, \lambda = 0$	N	-0.000008	$[-0.000016, -1.85 \times 10^{-7}]$	0.000168	negative †
$c = 0, \lambda = 0$	L	-0.000031	$[-0.000043, -0.000020]$	0.000168	negative †
$c = 0, \lambda = 0.5$	G	0.489453	[0.481962, 0.496944]	0.000168	practical +
$c = 0, \lambda = 0.5$	N	0.000944	[0.000088, 0.001799]	0.000168	positive
$c = 0, \lambda = 0.5$	L	0.010616	[0.010033, 0.011200]	0.000168	practical +
$c = 0, \lambda = 1$	G	0.978026	[0.962552, 0.993500]	0.000168	practical +
$c = 0, \lambda = 1$	N	0.001938	[0.000255, 0.003621]	0.000168	practical +
$c = 0, \lambda = 1$	L	0.021287	[0.020199, 0.022376]	0.000168	practical +
$c = 1, \lambda = 0$	G	0.001761	$[-0.071260, 0.074781]$	0.033542	unresolved
$c = 1, \lambda = 0$	N	0.069128	[0.028663, 0.109593]	0.033542	positive
$c = 1, \lambda = 0$	L	0.263144	[0.217187, 0.309101]	0.033542	practical +
$c = 1, \lambda = 0.5$	G	-0.004395	$[-0.043672, 0.034882]$	0.016863	unresolved
$c = 1, \lambda = 0.5$	N	0.040663	[0.019789, 0.061538]	0.016863	practical +
$c = 1, \lambda = 0.5$	L	0.154710	[0.122867, 0.186552]	0.016863	practical +
$c = 1, \lambda = 1$	G	0.011324	[0.000882, 0.021767]	0.000168	practical +
$c = 1, \lambda = 1$	N	0.012111	[0.005647, 0.018574]	0.000168	practical +
$c = 1, \lambda = 1$	L	0.046854	[0.040400, 0.053307]	0.000168	practical +
Nonlinear AR	G	-1.89×10^{-7}	$[-0.000050, 0.000049]$	0.000253	unresolved †
Nonlinear AR	N	-0.000016	$[-0.000062, 0.000029]$	0.000253	unresolved †
Nonlinear AR	L	0.000433	[0.000368, 0.000498]	0.000253	practical +
Van der Pol	G	-0.049419	$[-0.060532, -0.038307]$	0.002036	negative
Van der Pol	N	0.013443	[0.005084, 0.021803]	0.002036	practical +
Van der Pol	L	0.096633	[0.092436, 0.100830]	0.002036	practical +
Duffing	G	-0.030948	$[-0.035414, -0.026482]$	0.000338	negative
Duffing	N	0.005558	[0.001721, 0.009395]	0.000338	practical +
Duffing	L	0.046168	[0.041729, 0.050608]	0.000338	practical +
$c = 0$ interaction	I_0	-0.978039	$[-0.993512, -0.962566]$	0.000168	opposite sign
$c = 1$ interaction	I_1	-0.009564	$[-0.082839, 0.063712]$	0.016855	unresolved

†: the entire interval lies within the fixed practical band $[-\delta, \delta]$. An unresolved sign alone does not establish equivalence.

Table 17: Complete fresh capacity family (12 contrasts). All numerical loss quantities are multiplied by 10^3. Intervals use the two-sided 95% Bonferroni paired-replicate Student procedure with 23 degrees of freedom. Under paired normality, each t interval is exact and Bonferroni family coverage is at least 95%; otherwise coverage is approximate. The two stages have separate families.

Setting / budget	Contrast	$\overline{D}$	Simultaneous interval	δ	Assessment
Nonlinear AR / 64	Raw–K	-0.4556	$[-0.5235, -0.3878]$	0.2201	negative
Nonlinear AR / 64	Affine–K	0.0325	$[-0.0031, 0.0680]$	0.2201	unresolved †
Nonlinear AR / 256	Raw–K	-0.4791	$[-0.6013, -0.3569]$	0.2201	negative
Nonlinear AR / 256	Affine–K	0.0598	$[-0.0131, 0.1327]$	0.2201	unresolved †
Van der Pol / 64	Raw–K	1.8320	$[1.7369, 1.9271]$	0.1489	practical +
Van der Pol / 64	Affine–K	0.3501	$[0.2755, 0.4247]$	0.1489	practical +
Van der Pol / 256	Raw–K	0.7822	$[0.7550, 0.8094]$	0.1489	practical +
Van der Pol / 256	Affine–K	0.2060	$[0.1810, 0.2311]$	0.1489	practical +
Duffing / 64	Raw–K	-0.7557	$[-1.0831, -0.4282]$	0.1727	negative
Duffing / 64	Affine–K	-1.1605	$[-1.4707, -0.8503]$	0.1727	negative
Duffing / 256	Raw–K	-1.2372	$[-1.5242, -0.9503]$	0.1727	negative
Duffing / 256	Affine–K	-1.2756	$[-1.5702, -0.9810]$	0.1727	negative

†: the entire interval lies within the fixed practical band $[-\delta, \delta]$. An unresolved sign alone does not establish equivalence.

E Representations and fitted forecasting rules

A history $x \in \mathbb{R}^d$ contains the observations available at a forecast origin; its response $y \in \mathbb{R}^H$ contains the next H observations. For the univariate tasks below, the input dimension d equals the history length L. We fit direct multioutput regressors to historical pairs $\{(x_i, y_i)\}_{i=1}^n$. The construction below specifies which part of the regressor uses KAHM geometry. It does not assume that overlapping forecasting windows are independent. All input transformations, reference sets, and decoder bases are fitted on the training portion of each fit.

E.1 Shared reference geometry

Write $z(x) = D^{-1}(x - \bar{x})$, where $\bar{x}$ and the diagonal scales D are training means and population standard deviations. A standard deviation at most 10^{-12} is replaced by one. Thus this transformation retains every raw coordinate, including coordinates constant in the observed training set. Three independently seeded k-means partitions, with $k \in \{2, 4, 8\}$, use the same seeded subset of at most 512 training histories. Each partition has five initializations and at most 150 iterations. Each cluster supplies at most 24 references sampled without replacement; all its rows are used if there are fewer. For a common seed and training set, the alternative gates use identical partition and reference-selection rules. They are nevertheless fitted as separate computational pipelines.

Index the 14 local models by c, with the partition blocks denoted $\mathcal{C}_b$. Let $Q_c = [q_{c1}, \ldots, q_{cm_c}] \in \mathbb{R}^{d \times m_c}$ contain the training-standardized references, stored after the inherited float32 conversion, and let $\mu_c = m_c^{-1} Q_c \mathbf{1}$. The orthonormal columns of $P_c \in \mathbb{R}^{d \times r_c}$ are leading eigenvectors of the centered reference covariance. The requested rank is $\min(2, d-1, m_c - 1)$; only eigenvalues above double-precision machine epsilon are retained. The inherited implementation further reduces a rank greater than one if the smallest projected coordinate range is below 10^{-3}. A reference set with no usable direction has $r_c = 0$ and uses its center as its reconstruction. Other numerical failures remain recorded failures.

The affine reconstruction and centered local coordinates are

$$a_c(z) = \mu_c + P_c P_c^\top (z - \mu_c), \qquad u_c(z) = P_c^\top (z - \mu_c). \tag{9}$$

The affine and locality controls compute these quantities without fitting or querying any KAHM kernel.

E.2 Reconstruction scores and local gates

KAHM reconstruction. Kernel Affine Hull Machines construct nonlinear reconstructions from local kernel weights (Kumar et al., 2024). We retain the pinned implementation's local autoencoder, rather than asserting equivalence to an independently optimized end-to-end KAHM learner. For $r_c > 0$, let Σ_c be the sample covariance of the columns of $P_c^\top Q_c$. Its Gaussian kernel is

$$k_c(z,q) = \exp\left\{-\frac{(P_c^\top(z-q))^\top \Sigma_c^{-1} P_c^\top (z-q)}{2r_c}\right\}, \qquad (K_c)_{ij} = k_c(q_{ci}, q_{cj}). \tag{10}$$

To specify the inherited local regularizer exactly, put $\nu_c = \|Q_c\|_F^2/(dm_c)$ and initialize $\lambda_0 = 5\nu_c/2$. Its iteration is

$$B_t = (K_c + \lambda_t I)^{-1}, \qquad \lambda_{t+1} = 2\nu_c + \frac{\|Q_c - Q_c(K_c B_t)^\top\|_F^2}{dm_c}. \tag{11}$$

It stops after an update of size at most 10^{-3} or 100 updates, and retains the last computed B_t as B_c. In particular, it does not perform an extra inverse at the newly updated λ_{t+1}. For $k_c(z) = (k_c(z, q_{cj}))_{j=1}^{m_c}$, define

$$v_c(z) = B_c^\top k_c(z), \qquad \widehat{z}_c(z) = \frac{Q_c v_c(z)}{\mathbf{1}^\top v_c(z)}. \tag{12}$$

The weights may be signed. The reconstruction is therefore not generally a convex average. The inherited exact-zero-denominator convention replaces the denominator by one; that convention is an implementation choice, not a definition of a continuous extension at a singularity.

Stable Gaussian evaluation. For queries we compute the analytic log kernel $\ell_{cj}(z) = -\|S_c P_c^\top (z - q_{cj})\|^2/(2r_c)$, where S_c is the stored symmetric square root of the inverse covariance, and use $\widetilde{k}_{cj}(z) = \exp\{\ell_{cj}(z) - \max_j \ell_{cj}(z)\}$ in Equation (12). In exact arithmetic, $S_c^\top S_c = \Sigma_c^{-1}$, and multiplying every kernel entry for one query by the same positive constant cancels from the normalized reconstruction whenever its denominator is nonzero. This calculation avoids underflow caused solely by a large common negative log-kernel offset. It does not prevent signed cancellation, guarantee a well-conditioned denominator, or guarantee equivalence at an exact zero. The implementation retains the zero-denominator convention, records numerical diagnostics, and rejects nonfinite outputs. The historical mode instead exponentiates the unshifted logs through the inherited query routine; it is retained for equivalence checks, while this study uses the shifted evaluation.

Matched folding gates. Given either the KAHM reconstruction $\widehat{z}_c$ or the affine reconstruction a_c, denote it by $r_c^{\mathrm{rec}}(z)$. Define

$$\begin{aligned} \rho_c(z) &= 1 - \exp\{-\|z - r_c^{\mathrm{rec}}(z)\|\}, \qquad \theta_c(z) = \frac{1}{\pi}\arccos\left(\mathrm{clip}_{[-1,1]}\,\frac{z^\top r_c^{\mathrm{rec}}(z)}{\|z\|\,\|r_c^{\mathrm{rec}}(z)\|}\right), \\ s_c(z) &= \sqrt{\{\rho_c(z)^2 + \theta_c(z)^2\}/2}, \qquad \phi_c(z) = \frac{[1 - \mathrm{clip}_{[0,1]}\, s_c(z) + \tau]^\omega}{\sum_{j\in\mathcal{C}_b}[1 - \mathrm{clip}_{[0,1]}\, s_j(z) + \tau]^\omega}, \quad c \in \mathcal{C}_b. \end{aligned} \tag{13}$$

When either norm in the cosine denominator is zero, its cosine is set to zero, so that $\theta_c = 1/2$. The fixed values are $\omega = 16$ and $\tau = 10^{-6}$; normalization is evaluated with log-sum-exp. The comparison between these two gates changes the reconstruction while retaining the score, reference geometry, and local-coordinate construction.

Locality control. The third gate uses the historical native-affine score formula, applied to the same references. With $\epsilon = 10^{-6}$, set

$$t_c = \max\left\{m_c^{-1}\sum_j \|q_{cj} - \mu_c\|^2, \epsilon\right\},$$

$$
\begin{aligned}
e_c &= \max\left\{ m_c^{-1} \sum_j \|q_{cj} - a_c(q_{cj})\|^2, 0.05 t_c, \epsilon \right\}, \\
b_c(z) &= \frac{\|z - a_c(z)\|^2}{e_c} + 0.1 \frac{\|z - \mu_c\|^2}{t_c}, \\
\phi_c(z) &= \frac{\exp[-b_c(z)]}{\sum_{j \in \mathcal{C}_b} \exp[-b_j(z)]}.
\end{aligned} \tag{14}
$$

This distinguishes proximity to the local center from projection residual. Here "native" identifies a score formula; it does not identify a separately tuned native learning algorithm. It supplies a non-KAHM locality alternative to the matched folding-score comparison.

E.3 Local codes and finite forecasting heads

For each gate family g, concatenate all gates and gated local coordinates:

$$
T_g(x) = \big((\phi_c(z(x)))_c, (\phi_c(z(x))u_c(z(x)))_c\big), \qquad T_{\mathrm{raw}+g}(x) = (z(x), T_g(x)). \tag{15}
$$

The stored local width is $14 + \sum_c r_c \le 42$; the actual width is recorded. The raw augmentation prepends all d history coordinates, with no truncation when $d > 28$. The combination of local association and local position is related to coordinate and tangent coding (Yu et al., 2009; Yu & Zhang, 2010); our gates and decoder are the explicit constructions above. The stack of $P_c^\top$ has rank at most 28. A deficient stack alone does not prove information loss in T_g, because the gates also depend on the query. Raw augmentation gives the simpler unconditional reconstruction property stated below, without a rank assumption.

Nyström plus linear basis. Let $v(x)$ denote $z(x)$, $T_g(x)$, or $T_{\mathrm{raw}+g}(x)$, and let $q(x)$ be its coordinatewise training standardization. Select M training rows without replacement as landmarks $a_1, \ldots, a_M$ in q-space. The requested M is 32 or 64, capped by the training size. Set σ^2 to the median positive squared inter-landmark distance and define

$$
k_\sigma(q, a) = \exp\{-\|q - a\|^2/(2\sigma^2)\}, \qquad b(x) = \Big(q(x),\ \mathrm{diag}(\eta_J^{-1/2}) U_J^\top (k_\sigma(q(x), a_j))_{j=1}^M\Big). \tag{16}
$$

Here $U \,\mathrm{diag}(\eta) U^\top$ diagonalizes the landmark Gram matrix, and J retains eigenvalues exceeding $\max(10^{-10}\eta_{\max}, 10^{-12})$ (Williams & Seeger, 2001). Distances at most 10^{-14} are excluded from the bandwidth median; if none remain the squared bandwidth is one, and in all cases it is floored at 10^{-12}. The bandwidth multiplier is fixed to one. Finally, standardize b on training rows to obtain $f(x)$ and the design $F_{i,:} = f(x_i)^\top$. Both head standardizations replace scales below 10^{-8} by one. Raw ridge omits the Nyström block and uses the same centering and scaling convention.

For each horizon h, the fitted decoder solves

$$
\min_{\beta_h, a_h} \ \frac{1}{n} \sum_{i=1}^n [y_{ih} - a_h - f(x_i)^\top \beta_h]^2 + \alpha \|\beta_h\|^2. \tag{17}
$$

Thus the normal equations use $F^\top F/n + \alpha I$, and the intercept is unpenalized. Internally, targets are standardized using training statistics and transformed back after prediction. For fixed α, linearity makes this exactly equivalent in real arithmetic to Equation (17) in the original target units. This is a Euclidean ridge penalty on a standardized finite basis, rather than the canonical Gaussian RKHS norm. Adding local features also changes the Gaussian distances and the resulting Nyström basis; the raw and augmented nonlinear decoder classes need not be nested.

One common α from the ordered grid $(0.01, 1, 100)$ is selected by validation MSE averaged over origins and horizons in original target units, with the first grid entry winning ties. Within a method, geometry, training and validation features, basis construction, and the Gram eigensystem are computed once; all three alphas reuse those quantities. Different method variants share no fitted bank or free preprocessing. The selected configuration is then fitted anew on the admissible pretest training data. Search designs and eigensystems are not retained in the serialized final model. This gives an explicit efficient search whose full computation can be measured.

E.4 Information and finite-decoder risk

The following elementary facts delimit the interpretation of the experiments. They are established information-preservation and linear-smoother algebra, not new KAHM approximation theorems or performance guarantees.

Proposition 8 (Information preserved by raw augmentation)**.** *Condition on a fitted state Θ containing $\bar{x}$, positive diagonal D, and a measurable, finite-valued feature map T_g on the support of X. Then $T_{\mathrm{raw}+g}(X)$ and X generate the same information conditional on Θ. For a square-integrable response Y, their conditional expectations and optimal squared-loss Bayes risks coincide, conditional on Θ. This statement requires neither positive gates nor a full-rank stack of local bases.*

Proof. The augmented code is a measurable function of (X, Θ), and its first d coordinates recover $X = \bar{x} + Dz(X)$. Consequently $\sigma(T_{\mathrm{raw}+g}(X), \Theta) = \sigma(X, \Theta)$. Conditional expectation onto these equal sigma-fields, and hence the minimum squared prediction error, is identical. □

This is an exact-arithmetic statement about the representation, not a conditioning guarantee or a claim that a particular fitted decoder uses every retained coordinate. Empirical gains from raw augmentation must therefore be explained through finite fitting, representation-dependent regularization, or numerical behavior, rather than increased Bayes information relative to the same raw input.

Proposition 9 (Fixed-feature, fixed-alpha risk identity)**.** *Consider one response coordinate, a fixed $\alpha > 0$, and the centered training design F above. Let $\mathcal{G}$ contain the training and query inputs and the randomness used to fit their unsupervised feature map, so F and $f(x_*)$ are $\mathcal{G}$-measurable. Put*

$$w_* = \frac{\mathbf{1}}{n} + F(F^\top F + n\alpha I)^{-1} f(x_*). \tag{18}$$

Let the training response vector be Y, the future response be Y_*, and assume their conditional second moments are finite. Define $\mu = \mathbb{E}[Y \mid \mathcal{G}]$, $\mu_* = \mathbb{E}[Y_* \mid \mathcal{G}]$, $\Sigma = \mathrm{Cov}(Y \mid \mathcal{G})$, $c = \mathrm{Cov}(Y, Y_* \mid \mathcal{G})$, and $v_* = \mathrm{Var}(Y_* \mid \mathcal{G})$. The fitted prediction is $\widehat{Y}_* = w_*^\top Y$, and

$$\mathbb{E}[(\widehat{Y}_* - Y_*)^2 \mid \mathcal{G}] = (w_*^\top \mu - \mu_*)^2 + w_*^\top \Sigma w_* + v_* - 2w_*^\top c. \tag{19}$$

Proof. Because $F^\top \mathbf{1} = 0$, the unpenalized intercept is $\bar{Y}$ and $\widehat{\beta} = (F^\top F + n\alpha I)^{-1} F^\top Y$. Substituting into $\bar{Y} + f(x_*)^\top \widehat{\beta}$ gives Equation (18); the inverse target standardization cancels as described above. Conditional on $\mathcal{G}$, expand the second moment of $w_*^\top Y - Y_*$ as its squared mean plus its variance. The covariance cross term is $-2w_*^\top c$, giving the identity. □

The identity holds without independent training responses or an independent future response. Overlapping windows and serial dependence can affect both Σ and c; neither may be discarded without additional assumptions. Averaging the identity over horizons gives the corresponding average risk. Changing geometry changes w_* and can change all its bias and covariance terms, without a universal ordering between representations. The identity does not automatically describe the full validation-selected procedure: after alpha selection, the smoother depends on observed validation responses. No generalization or selection guarantee is inferred from it.

F Natural-data protocol

Source allocation, task roster, model grids, and analysis are fixed before comparative outcomes. We test matched representations, practical accuracy–cost tradeoffs, and prospective selection on reserved families. Specifications, checksums, and execution freezes accompany the code as internal reproducibility records, not externally timestamped preregistration.

Table 18: Fixed source allocation and forecast regimes. "Selected/all" counts series, not independent source families. Each selected series uses both regimes for its frequency. L is history length, H forecast horizon, and s the declared seasonal period.

Phase	Source snapshot	Frequency	Selected/all
Development	Electricity (Godahewa et al., 2020a)	weekly	4/321
	NN5, with missing values (Godahewa et al., 2020c)	daily	4/111
	Tourism (Godahewa et al., 2020e)	monthly	4/366
	FRED-MD (Godahewa et al., 2020h)	monthly	4/107
Confirmation	Traffic (Godahewa et al., 2020f)	weekly	4/862
	Hospital (Godahewa et al., 2020b)	monthly	4/767
	Saugeen River flow (Godahewa et al., 2020d)	daily	1/1
	US births (Godahewa et al., 2020g)	daily	1/1

Frequency	Short (L, H)	Long (L, H)	s
Monthly	$(12, 3)$	$(24, 6)$	12
Weekly	$(13, 4)$	$(26, 8)$	52
Daily	$(28, 7)$	$(56, 14)$	7

F.1 Sources and chronological tasks

Table 18 gives eight versioned Monash snapshots (Godahewa et al., 2021), chosen by provenance, application diversity, length, and size. Related series, channels, and aggregations share a source family and cannot cross the boundary between development and confirmation. Models are local to each series, including differently dated Tourism series. The supplement retains the records' CC BY 4.0 licenses, versions, attribution, original-source references, and archive hashes.

Training-prefix eligibility requires at least 16 finite windows in each regime and a positive training error scale. A fixed salted hash of source and series ID orders eligible series; we retain the first four, or all if fewer exist, across regimes and capacities. No later loss or availability criterion selects replacements. The 26 series give 52 base tasks; two capacities and seeds 17 and 43 give 208 runs: 128 development and 80 confirmation, with four source families per phase.

For a length-n series indexed from zero, let $a = \lfloor 3n/5 \rfloor$ and $b = \lfloor 4n/5 \rfloor$. Training, validation, and test intervals are $[0, a)$, $[a, b)$, and $[b, n)$. An origin t is the last observed index, with

$$x_t = (y_{t-L+1}, \ldots, y_t), \qquad y_t^+ = (y_{t+1}, \ldots, y_{t+H}). \tag{20}$$

The complete target interval must lie within its split; contexts may include earlier observed values. Stateful controls receive $t + 1$, their first-future index. We retain raw snapshot values, including NN5 missing markers, and discard windows with an unavailable context or target without interpolation. Evenly spaced indices among available origins give at most 64 validation and 64 test windows, and the training cap in Table 19. Every method receives the same evaluation origins.

Before outcomes, length checks reduced the proposed monthly $(48, 6)$ and weekly $(52, 8)$ regimes, which yield zero Hospital and three Traffic training windows. The adopted long regimes yield 21 and 29, respectively. Context and horizon are paired, so their separate effects are not identified. NN5 T88 at $(56, 14)$ retains only seven test origins out of 146 candidates. All task availability counts are recorded.

All candidate preprocessing is fitted on training data, and validation MSE selects hyperparameters. The selected pipeline is then fitted anew on all eligible windows with targets preceding b, subject to its training cap. This refit legitimately includes windows crossing the former training/validation boundary. Parameters are fixed thereafter; earlier realized observations may enter later contexts or state updates. Test outcomes never retune parameters.

Table 19: Capacity caps. These are limits on examples and model size, not equal realized time budgets. A shorter source uses its available examples.

Setting	Training windows	RBF landmarks	Tree leaves/iterations	MLP width/iterations
Small	512	32	15/100	32/200
Medium	1,024	64	31/150	64/300

F.2 Comparators, capacity, and loss

The twelve methods comprise raw ridge and raw RBF heads; affine-, locality-, and KAHM-gated local RBF pipelines; each local pipeline augmented by the full raw history; and seasonal naive, Holt, histogram gradient boosting, and a one-hidden-layer MLP. Section E defines the feature maps and finite RBF head. The primary matched pair is raw RBF (R) versus raw-history plus KAHM RBF (K). The raw-history controls retain the inputs independently of local-basis rank (Proposition 8). All twelve methods remain in the declared practical portfolio.

The geometric bank uses cluster counts $(2, 4, 8)$, requested rank two, at most 24 references per local model, a 512-row fitting cap, $\omega = 16$, and $\tau = 10^{-6}$. Reduced effective ranks and numerical fallbacks are recorded. Each alternative constructs its own representation; affine/locality controls do not fit a KAHM kernel. Within a method, the fixed representation and head statistics are reused across ridge penalties $\{0.01, 1, 100\}$; the bandwidth multiplier is one. Boosting/MLP regularization is chosen from $\{0.01, 1\}$, Holt selects damped or undamped trend, and seasonal naive has no tuned parameter. These modest declared grids do not imply optimal tuning of every comparator.

Seasonal naive can use its declared lag from the observed past even when it exceeds L, using the last finite value for an unavailable lag and period one before a full seasonal lag exists. Holt fits a capped historical segment, forward-fills only within that segment, and assimilates observations strictly before each forecast; an absent online observation causes a state prediction without measurement update. Boosting and MLP fit direct multi-output forecasts without a random internal validation split.

For recorded origins $\mathcal{O}$, the task score is

$$\ell = \frac{1}{|\mathcal{O}|Hq} \sum_{t \in \mathcal{O}} \sum_{h=1}^{H} (\widehat{y}_{t+h} - y_{t+h})^2, \tag{21}$$

Here q is the seasonal-difference mean square in the uncapped initial training prefix, requiring two finite pairs and positive finite q. First differences provide the fallback; a series with neither scale is ineligible. This denominator remains fixed during refitting. Raw MSE, q, predictions, and origin losses are saved and independently checked.

Aggregation gives equal mass to families and then each nested dataset, series, horizon, history, capacity, and seed. These nested observations are not independent source replications. We report family outcomes, dispersion, and leave-one-family-out sensitivity, without broad-population confidence intervals or window-level significance claims.

F.3 Measured pipeline and selector costs

Methods execute sequentially in fresh child processes with one BLAS thread and deterministically shuffled order, limited to 120 seconds per method and 1,800 per task. Wall and process CPU time are separate. Each warm-library scientific pipeline includes data/checksum preparation, array loading, configuration fitting and validation, selection bookkeeping, selected refitting, forecasting, and model serialization. Source preparation is equally allocated across its tasks and charged consistently to alternatives. Shared imports are initialized before measurement; initialization, process launch/IPC, and launch-to-result duration are recorded separately rather than allocated arbitrarily to policies.

Let S_m contain method m's search nodes and F_m its final refit/forecast nodes, including serialization. A validation selector over candidates $\mathcal{A}$, choosing m^*, requires

$$\mathcal{D}_{\text{val}} = \mathcal{D}_{\text{data}} \cup \bigcup_{m \in \mathcal{A}} S_m \cup F_{m^*}, \qquad C = \sum_{v \in \mathcal{D}_{\text{val}}} c_v + c_{\text{decision}}. \tag{22}$$

Set unions count shared dependencies once. A learned selector instead pays for its required metadata/geometry/probe nodes plus $S_{m^*} \cup F_{m^*}$, even when it rejects KAHM. Geometry-only acquisition is conservatively billed through its measured fitted-probe component. Thus policy costs are counterfactual sums of measured required computations, not the cost of running every alternative in the benchmark.

Development-label acquisition also costs the R, K searches, refits, and forecasts, plus each variant's feature/probe nodes. We report this investment separately from regression fitting, online policy costs, and the full research benchmark's elapsed time, without assuming a deployment count for amortization. Summed standalone task costs are conservative where sharing across seeds or capacities could reduce work.

We also record model bytes and process high-water RSS. RSS includes the interpreter, inherited arrays, and earlier process history; it is neither incremental memory nor an additive exact policy-memory reconstruction. No GPU measurement is inferred from CPU timing. Failed attempts remain visible and retain the available timing records. A failed method cannot be replaced or omitted silently; analysis requires the complete declared task/method roster.

F.4 Prospective diagnostic and confirmation gate

The diagnostic predicts benefit before a task's test outcomes, while using its training history and permitted validation information. Its development label is $\log[(\ell_R + \epsilon)/(\ell_K + \epsilon)]$, with $\epsilon = 10^{-12}$. Development test errors may train this cross-task rule; confirmation errors cannot train, standardize, tune, or select it.

We retain four weighted ridge rules: metadata M, M plus geometry G, M plus a validation probe P, and $M + P + G$. Metadata features are $\log(1 + N_{\text{train}})$, $\log(1 + H/L)$, and training-prefix lag-one autocorrelation. Geometry adds normalized KAHM gate entropy and $\log(1 + r)$, where r is the gated affine-projection residual on probe training inputs. The probe fits the matched R, K pair with at most 128 training windows, 16 landmarks, a 128-row bank cap, and penalty one; its feature is the log ratio of scaled validation losses with the same loss floor.

The diagnostic penalty is fixed at one, with an unpenalized intercept. Source-balanced feature standardization and fitting occur inside each development fold, holding out an entire source family. All four variants are then fitted on all development families and frozen before confirmation. The primary incremental comparison is $M + P + G$ against $M + P$, testing whether geometry adds value beyond metadata and a reduced-capacity validation probe. Controls include always R, always K, direct probe selection, validation selection within the matched pair, the six-method raw/practical portfolio, the two-method KAHM portfolio, and the full twelve-method portfolio.

A positive predicted log benefit selects K; ties select R. A fixed 5% relative MSE change defines secondary practical benefit/harm, not a tuned selection threshold. All four variants are reported without choosing one from confirmation performance. A test-set oracle is only a hindsight regret reference. Utility requires improved held-out loss–cost behavior against metadata and probe controls, not merely correlation with retrospective gains.

Before confirmation, saved development predictions are audited against frozen targets and reported errors. The final freeze binds code, dependencies, roster, development records, and diagnostic coefficients. Confirmation cannot run an arbitrary subset; only complete checksum-verified results may be resumed. This is a selected, small collection, not a random sample of forecasting domains or a state-of-the-art survey. Seasonal periods are conventions rather than fitted optimal cycles. In particular, FRED-MD is a fixed-vintage snapshot: its metadata describe transformed series without a complete per-series preprocessing audit (Godahewa et al., 2020h). Our train-only guarantees apply to processing in this extension, not to unverified upstream decisions, historical revisions, or release-time availability. The conclusions must remain conditional on these sources, contexts, grids, and resource settings.

Table 20: Standalone confirmation methods, equally weighted across 4 original source families using the frozen within-source hierarchy. Loss is training-scale-normalized MSE. Wall and CPU seconds cover complete warm-library scientific pipelines. RSS is the averaged maximum recorded process high-water mark, not incremental memory; model size is serialized bytes in KiB. No uncertainty intervals are inferred from dependent tasks.

Method	Scaled MSE	Wall (s)	CPU (s)	RSS (MiB)	Model (KiB)
Raw ridge	0.7798	0.188	0.1879	171.5	5.655
Raw RBF	0.8075	0.1688	0.1687	171.5	37.32
Affine local	0.9449	0.5002	0.4897	171.5	93.2
Locality local	0.8964	0.2898	0.2898	171.5	93.4
KAHM local	0.9354	0.3218	0.3214	171.5	164.6
Raw + affine	0.8458	0.3379	0.3265	171.5	107.8
Raw + locality	0.8462	0.2791	0.279	171.5	107.8
Raw + KAHM	0.8419	0.3874	0.3814	171.5	179.2
Seasonal naive	1.05	0.1614	0.1613	171.5	0.1382
Holt	2.373	0.3449	0.3446	171.5	0.4219
Histogram boosting	0.9555	8.91	8.901	171.5	2591
MLP	0.8785	1.463	1.462	171.5	71.62

Table 21: Confirmation cost transparency for the primary matched pair, with equal source-family weighting. Total is the complete warm-library scientific pipeline. Cleanup is its measured selection-overhead component, including explicit garbage collection; the percentage is the ratio of aggregated component and total times. Launch separately reports the measured isolated-child launch-to-result interval. These are implementation-specific costs, not intrinsic kernel arithmetic or universal complexity ratios.

Method	Total (s)	Cleanup (s)	Cleanup (%)	Launch (s)
Raw RBF	0.16879	0.13808	81.8	0.17748
Raw + KAHM	0.38741	0.18253	47.11	0.40097

G Additional natural-source results and diagnostics

The following tables retain all declared methods, selection policies, source means, capacities, diagnostic variants, and recorded fitting warnings. These are descriptive summaries under the same fixed source hierarchy. They do not pool the natural-source results with the controlled-study confidence families. Full precision, forecasts, targets, origins, selected parameters, and cost components are retained in the anonymous supplement.

Table 22: Frozen-policy confirmation outcomes over 4 source families with the same hierarchical weighting. Gain is raw-RBF scaled MSE minus chosen loss. Harm is the weighted fraction with loss more than 5% above raw RBF. The final column counts selection of the specific raw-plus-KAHM pair member, including in portfolios. Wall cost includes all required probes/searches and decision inference, then the chosen final refit, forecast and model serialization. Offline evidence acquisition and diagnostic fitting are separate.

Policy	Scaled MSE	Wall (s)	Gain	Harm (%)	Raw+K (%)
Always raw	0.8075	0.1688	0	0	0
Always raw + KAHM	0.8419	0.3874	-0.0344	25	100
Validation: matched pair	0.8097	0.4812	-0.002159	5.47	32
Validation: raw portfolio	0.8639	8.338	-0.05638	10.9	0
Validation: KAHM family	0.8406	0.6242	-0.03306	26.6	73.4
Validation: full portfolio	0.8764	9.948	-0.0689	14.8	14.1
Cheap probe	0.8143	0.3465	-0.006771	9.38	50
Metadata	0.816	0.2976	-0.008445	15.6	50
Metadata + geometry	0.816	0.3695	-0.008445	15.6	50
Metadata + probe	0.8071	0.3309	0.0004363	3.12	31.2
Metadata + probe + geometry	0.8071	0.3309	0.0004363	3.12	31.2

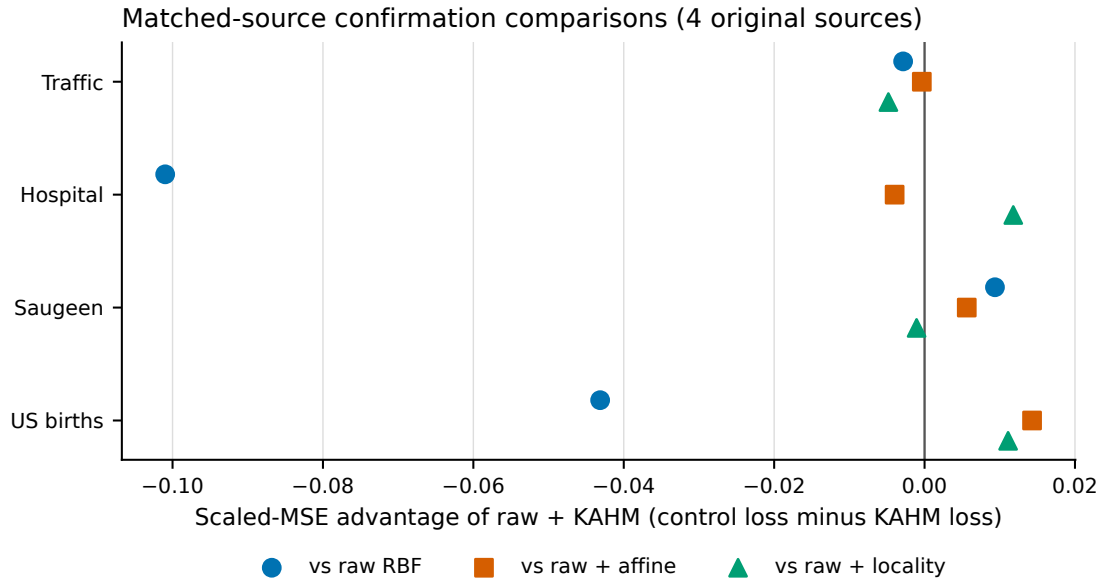


Figure 3: Within-source confirmation comparisons. Positive differences favor raw plus KAHM. Points compare the same hierarchically weighted source means; no window-level uncertainty intervals are inferred. The small KAHM differences against structural controls change sign across sources.

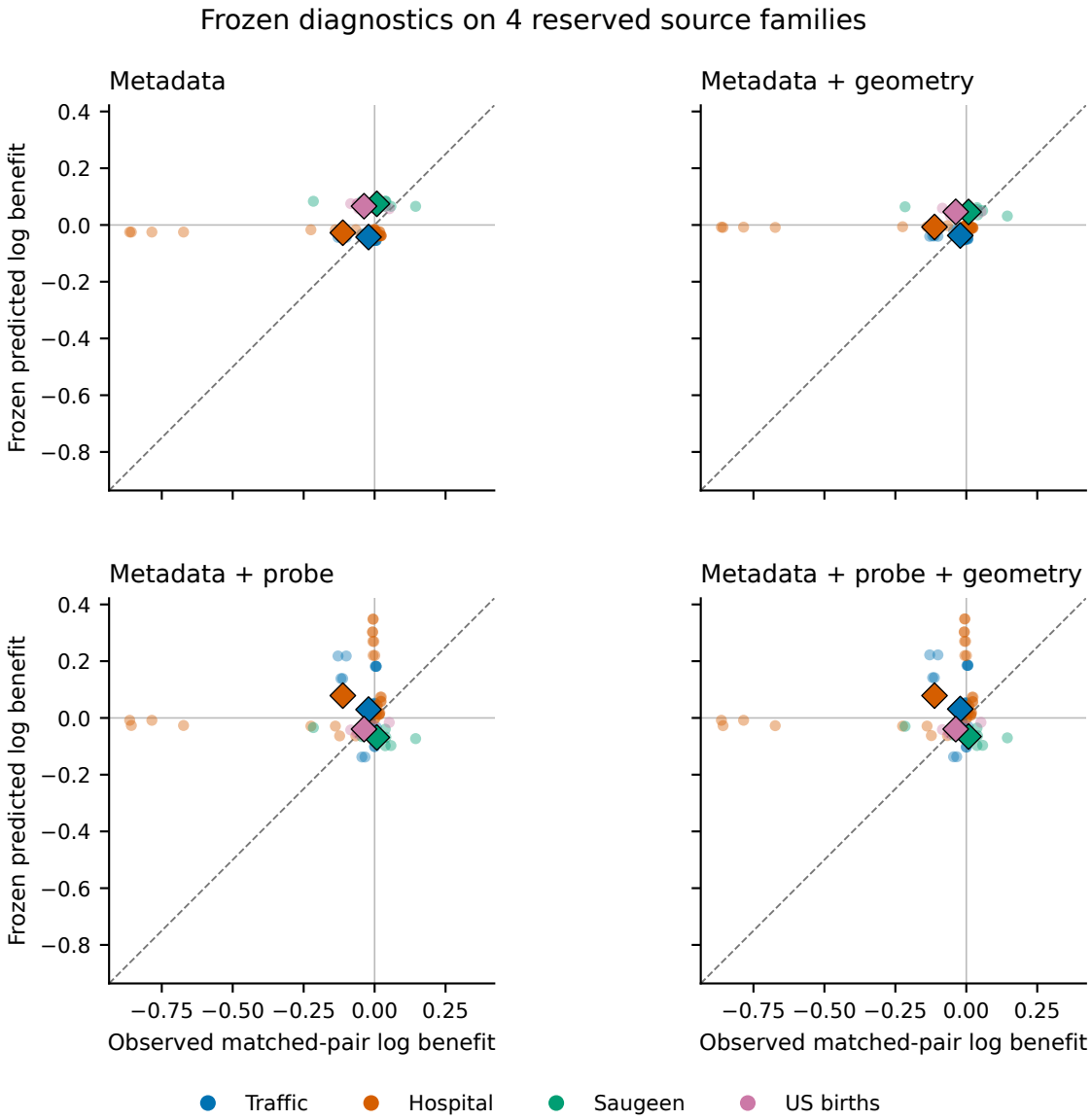


Figure 4: Observed versus frozen predicted log benefit. Points within a source are dependent task runs, and diamonds are source means. No recalibration is fitted on confirmation. Similar decisions can coexist with different estimated benefit magnitudes.

Table 23: Development scaled MSE by original source family. The final column gives equal-source aggregation; source columns retain the declared within-source weights. Development diagnostics use leave-one-source-family-out fits; confirmation diagnostics use the frozen development fit. Always-policy rows duplicate their corresponding standalone methods deliberately.

Method or policy	FRED-MD	NN5	Tourism	Electricity	Equal-source
Raw ridge	0.5601	0.8584	2.865	0.4243	1.177
Raw RBF	10.46	0.7552	3.476	0.4388	3.783
Affine local	2.548	0.8795	3.769	0.4535	1.912
Locality local	23.83	0.8148	3.778	0.4112	7.209
KAHM local	12.23	0.9163	3.636	0.4525	4.309
Raw + affine	1.409	0.7614	3.172	0.4481	1.448
Raw + locality	10.18	0.7474	3.046	0.4418	3.603
Raw + KAHM	4.254	0.7704	3.229	0.4539	2.177
Seasonal naive	2.269	1.089	2.982	0.5507	1.723
Holt	0.2409	1.358	8.228	0.6984	2.631
Histogram boosting	34.02	0.8032	4.786	0.4404	10.01
MLP	1.022	0.8475	2.77	0.4715	1.278
Always raw	10.46	0.7552	3.476	0.4388	3.783
Always raw + KAHM	4.254	0.7704	3.229	0.4539	2.177
Validation: matched pair	5.267	0.7335	3.229	0.4415	2.418
Validation: raw portfolio	0.2409	0.8021	3.014	0.4528	1.127
Validation: KAHM family	4.104	0.7792	3.223	0.4572	2.141
Validation: full portfolio	0.2409	0.8052	3.092	0.4675	1.152
Cheap probe	5.106	0.742	3.25	0.4217	2.38
Metadata	4.254	0.7704	3.229	0.4539	2.177
Metadata + geometry	4.254	0.7704	3.394	0.4539	2.218
Metadata + probe	4.157	0.759	3.338	0.4342	2.172
Metadata + probe + geometry	4.215	0.7704	3.338	0.4342	2.189

Table 24: Development warm-library wall seconds by original source family. The final column gives equal-source aggregation; source columns retain the declared within-source weights. Development diagnostics use leave-one-source-family-out fits; confirmation diagnostics use the frozen development fit. Always-policy rows duplicate their corresponding standalone methods deliberately.

Method or policy	FRED-MD	NN5	Tourism	Electricity	Equal-source
Raw ridge	0.3731	0.1631	0.1888	0.1873	0.2281
Raw RBF	0.1929	0.1912	0.1834	0.1692	0.1842
Affine local	0.3285	0.2805	0.2983	0.2863	0.2984
Locality local	0.5431	0.29	0.2784	0.2649	0.3441
KAHM local	0.3705	0.3696	0.3735	0.3369	0.3626
Raw + affine	0.2923	0.2998	0.2979	0.288	0.2945
Raw + locality	0.3185	0.2967	0.2988	0.2533	0.2918
Raw + KAHM	0.4275	0.3807	0.3252	0.3024	0.3589
Seasonal naive	0.2125	0.177	0.1834	0.1878	0.1902
Holt	0.3224	0.42	0.3682	0.316	0.3567
Histogram boosting	3.086	4.954	1.729	1.819	2.897
MLP	1.278	0.9551	0.8521	0.5508	0.9089
Always raw	0.1929	0.1912	0.1834	0.1692	0.1842
Always raw + KAHM	0.4275	0.3807	0.3252	0.3024	0.3589
Validation: matched pair	0.5562	0.5313	0.4929	0.4301	0.5026
Validation: raw portfolio	4.066	5.306	2.641	2.719	3.683

Method or policy	FRED-MD	NN5	Tourism	Electricity	Equal-source
Validation: KAHM family	0.692	0.6568	0.6115	0.5511	0.6278
Validation: full portfolio	5.861	6.513	4.138	4.075	5.147
Cheap probe	0.4739	0.4615	0.4004	0.3328	0.4172
Metadata	0.4284	0.3815	0.3258	0.3031	0.3597
Metadata + geometry	0.5195	0.465	0.3474	0.3744	0.4266
Metadata + probe	0.5029	0.474	0.2891	0.3411	0.4018
Metadata + probe + geometry	0.5284	0.4828	0.2891	0.341	0.4103

Table 25: Confirmation scaled MSE by original source family. The final column gives equal-source aggregation; source columns retain the declared within-source weights. Development diagnostics use leave-one-source-family-out fits; confirmation diagnostics use the frozen development fit. Always-policy rows duplicate their corresponding standalone methods deliberately.

Method or policy	Traffic	Hospital	Saugeen	US births	Equal-source
Raw ridge	0.5737	0.8461	0.522	1.177	0.7798
Raw RBF	0.5739	0.8228	0.6106	1.223	0.8075
Affine local	0.5796	0.928	0.6462	1.626	0.9449
Locality local	0.5719	0.7702	0.6504	1.593	0.8964
KAHM local	0.5775	0.9029	0.6652	1.596	0.9354
Raw + affine	0.5764	0.9198	0.6068	1.28	0.8458
Raw + locality	0.572	0.9356	0.6002	1.277	0.8462
Raw + KAHM	0.5768	0.9238	0.6012	1.266	0.8419
Seasonal naive	0.6484	0.7848	1.023	1.743	1.05
Holt	0.6842	1.003	0.6079	7.197	2.373
Histogram boosting	0.6524	1.025	0.5386	1.606	0.9555
MLP	0.6811	1.15	0.5365	1.147	0.8785
Always raw	0.5739	0.8228	0.6106	1.223	0.8075
Always raw + KAHM	0.5768	0.9238	0.6012	1.266	0.8419
Validation: matched pair	0.5738	0.8236	0.6042	1.237	0.8097
Validation: raw portfolio	0.6087	1.116	0.5639	1.167	0.8639
Validation: KAHM family	0.577	0.9182	0.6012	1.266	0.8406
Validation: full portfolio	0.6122	1.099	0.5695	1.225	0.8764
Cheap probe	0.5761	0.8211	0.6253	1.235	0.8143
Metadata	0.5739	0.8228	0.6012	1.266	0.816
Metadata + geometry	0.5739	0.8228	0.6012	1.266	0.816
Metadata + probe	0.5761	0.8189	0.6106	1.223	0.8071
Metadata + probe + geometry	0.5761	0.8189	0.6106	1.223	0.8071

Table 26: Confirmation warm-library wall seconds by original source family. The final column gives equal-source aggregation; source columns retain the declared within-source weights. Development diagnostics use leave-one-source-family-out fits; confirmation diagnostics use the frozen development fit. Always-policy rows duplicate their corresponding standalone methods deliberately.

Method or policy	Traffic	Hospital	Saugeen	US births	Equal-source
Raw ridge	0.1454	0.1959	0.1905	0.2204	0.188
Raw RBF	0.1644	0.1636	0.1693	0.1779	0.1688
Affine local	0.2424	0.2732	0.4038	1.082	0.5002
Locality local	0.2311	0.254	0.4021	0.2721	0.2898
KAHM local	0.2579	0.2764	0.3496	0.4035	0.3218

Method or policy	Traffic	Hospital	Saugeen	US births	Equal-source
Raw + affine	0.3969	0.2809	0.354	0.3199	0.3379
Raw + locality	0.2223	0.2752	0.3166	0.3022	0.2791
Raw + KAHM	0.2674	0.4235	0.4676	0.3911	0.3874
Seasonal naive	0.1551	0.164	0.1874	0.1393	0.1614
Holt	0.2759	0.3535	0.4118	0.3386	0.3449
Histogram boosting	1.774	1.39	17.59	14.89	8.91
MLP	0.4046	0.392	2.264	2.79	1.463
Always raw	0.1644	0.1636	0.1693	0.1779	0.1688
Always raw + KAHM	0.2674	0.4235	0.4676	0.3911	0.3874
Validation: matched pair	0.3909	0.5408	0.5273	0.4656	0.4812
Validation: raw portfolio	2.276	2.19	15.94	12.95	8.338
Validation: KAHM family	0.4622	0.6403	0.7116	0.6829	0.6242
Validation: full portfolio	3.595	3.661	17.63	14.9	9.948
Cheap probe	0.3029	0.2768	0.4534	0.3531	0.3465
Metadata	0.1651	0.1644	0.4686	0.3923	0.2976
Metadata + geometry	0.2216	0.2212	0.5797	0.4555	0.3695
Metadata + probe	0.2958	0.457	0.3033	0.2673	0.3309
Metadata + probe + geometry	0.2958	0.457	0.3033	0.2673	0.3309

Table 27: Development capacity comparison. Each capacity changes specified training-window and model-size caps; it is not an equal-time budget. Values are equal-source weighted and retain all prescribed context–horizon settings and seeds.

Method or policy	medium MSE	medium wall (s)	small MSE	small wall (s)
Raw ridge	1.177	0.28	1.177	0.1761
Raw RBF	4.582	0.1846	2.984	0.1838
Affine local	1.989	0.3292	1.836	0.2676
Locality local	8.238	0.4065	6.18	0.2817
KAHM local	3.855	0.3779	4.763	0.3473
Raw + affine	1.394	0.3071	1.501	0.282
Raw + locality	3.879	0.3038	3.327	0.2798
Raw + KAHM	2.332	0.3819	2.022	0.3359
Seasonal naive	1.723	0.2078	1.723	0.1725
Holt	2.631	0.345	2.632	0.3683
Histogram boosting	9.795	3.468	10.23	2.326
MLP	1.253	1.108	1.302	0.7098
Always raw	4.582	0.1846	2.984	0.1838
Always raw + KAHM	2.332	0.3819	2.022	0.3359
Validation: matched pair	2.342	0.5295	2.493	0.4758
Validation: raw portfolio	1.128	4.223	1.127	3.142
Validation: KAHM family	2.261	0.6594	2.021	0.5963
Validation: full portfolio	1.137	5.805	1.166	4.488
Cheap probe	2.297	0.4321	2.463	0.4022
Metadata	2.332	0.3828	2.022	0.3367
Metadata + geometry	2.364	0.4492	2.072	0.404
Metadata + probe	2.337	0.419	2.007	0.3846
Metadata + probe + geometry	2.348	0.4327	2.03	0.388

Table 28: Confirmation capacity comparison. Each capacity changes specified training-window and model-size caps; it is not an equal-time budget. Values are equal-source weighted and retain all prescribed context–horizon settings and seeds.

Method or policy	medium MSE	medium wall (s)	small MSE	small wall (s)
Raw ridge	0.7635	0.1561	0.7961	0.2199
Raw RBF	0.7767	0.1719	0.8383	0.1657
Affine local	0.916	0.3088	0.9738	0.6917
Locality local	0.8908	0.3176	0.902	0.2621
KAHM local	0.9243	0.334	0.9466	0.3097
Raw + affine	0.8233	0.3859	0.8684	0.2899
Raw + locality	0.8363	0.2831	0.8561	0.275
Raw + KAHM	0.8288	0.3732	0.8551	0.4016
Seasonal naive	1.05	0.1621	1.05	0.1607
Holt	2.373	0.333	2.373	0.3569
Histogram boosting	0.9215	12.73	0.9896	5.086
MLP	0.8712	1.956	0.8859	0.9695
Always raw	0.7767	0.1719	0.8383	0.1657
Always raw + KAHM	0.8288	0.3732	0.8551	0.4016
Validation: matched pair	0.7847	0.4538	0.8347	0.5085
Validation: raw portfolio	0.8404	11.6	0.8874	5.074
Validation: KAHM family	0.8281	0.6116	0.8531	0.6369
Validation: full portfolio	0.8541	12.98	0.8987	6.918
Cheap probe	0.7943	0.3749	0.8343	0.3182
Metadata	0.8009	0.3187	0.8311	0.2764
Metadata + geometry	0.8009	0.3779	0.8311	0.3611
Metadata + probe	0.7765	0.2805	0.8377	0.3812
Metadata + probe + geometry	0.7765	0.2805	0.8377	0.3812

Table 29: Estimated-benefit diagnostics with frozen hierarchical weights. These are descriptive log-benefit prediction errors and classification accuracies; they are not calibrated probabilities or interval coverage. No recalibration is fitted on confirmation data.

Diagnostic	Log RMSE	Log bias	Sign (%)	5% class (%)
Development: Metadata	1.035	0.01412	57	49.2
Development: Metadata + geometry	1.037	0.03491	47.7	43
Development: Metadata + probe	1.012	0.03064	52.3	53.9
Development: Metadata + probe + geometry	1.02	0.05091	50.8	43
Confirmation: Metadata	0.1643	0.05835	55.5	59.4
Confirmation: Metadata + geometry	0.1612	0.05269	55.5	75
Confirmation: Metadata + probe	0.1845	0.04069	58.6	73.4
Confirmation: Metadata + probe + geometry	0.1847	0.04196	58.6	71.9

Table 30: One-off diagnostic development costs. Acquisition sums the union of both matched predictors' complete search/final nodes and each variant's metadata/probe/geometry dependencies within every development task. Summing standalone tasks can conservatively repeat preparation across seeds/tasks. Acquire and total denote wall time; CPU covers acquisition. Do not add the four rows: their developmental labels and some features can be shared. Ridge fitting is measured separately; no deployment-count amortization is assumed.

Variant	Acquire (s)	CPU (s)	Ridge (ms)	Total (s)
Metadata	68.49	68.45	6.657	68.5
Metadata + geometry	78.6	78.56	10.04	78.61
Metadata + probe	80.83	80.79	9.035	80.84
Metadata + probe + geometry	80.83	80.79	6.146	80.84

Table 31: Frozen development-only standardization and ridge coefficients, rounded for display. G denotes geometry and P the cheap probe. An absent feature is shown by a dash. Predictions use the unrounded values in the frozen diagnostic JSON and the fixed zero decision threshold.

Feature	Mean	Scale	Meta	Meta+G	Meta+P	Meta+P+G
Intercept	–	–	0.023966	0.023966	0.023966	0.023966
log(1 + training windows)	5.1417	0.66192	0.017096	0.013047	-0.0099182	-0.0089504
log(1 + horizon/history)	0.23442	0.019538	-0.012299	-0.013178	-0.0043432	-0.0045441
Lag-one autocorrelation	0.7189	0.24263	0.0069415	0.001093	-0.014509	-0.012806
KAHM gate entropy	0.074168	0.063288	–	-0.010312	–	-0.0026401
log(1 + affine residual)	0.18213	0.12986	–	-0.013232	–	0.0063025
Probe log-loss ratio	0.25347	0.67921	–	–	0.14611	0.1467

Table 32: Recorded fitting behavior. Search and selected-refit warning events are separated. Counts describe task fits, which include dependent seeds and settings. Warned counts fits with recorded warnings; at cap counts selected refits reaching the recorded iteration limit. Full warning text is retained in the JSON report. No warning-triggered post hoc retuning is applied.

Method	Fits	Warned	Search	Refit	At cap
Development: Raw ridge	128	0	0	0	0
Development: Raw RBF	128	0	0	0	0
Development: Affine local	128	0	0	0	0
Development: Locality local	128	0	0	0	0
Development: KAHM local	128	0	0	0	0
Development: Raw + affine	128	0	0	0	0
Development: Raw + locality	128	0	0	0	0
Development: Raw + KAHM	128	0	0	0	0
Development: Seasonal naive	128	0	0	0	0
Development: Holt	128	0	0	0	0
Development: Histogram boosting	128	0	0	0	0
Development: MLP	128	128	232	105	105
Confirmation: Raw ridge	80	0	0	0	0
Confirmation: Raw RBF	80	0	0	0	0
Confirmation: Affine local	80	0	0	0	0
Confirmation: Locality local	80	0	0	0	0
Confirmation: KAHM local	80	0	0	0	0

Method	Fits	Warned	Search	Refit	At cap
Confirmation: Raw + affine	80	0	0	0	0
Confirmation: Raw + locality	80	0	0	0	0
Confirmation: Raw + KAHM	80	0	0	0	0
Confirmation: Seasonal naive	80	0	0	0	0
Confirmation: Holt	80	0	0	0	0
Confirmation: Histogram boosting	80	0	0	0	0
Confirmation: MLP	80	79	137	68	68

Table 33: Selection at numerical search-grid endpoints. Counts are descriptive; a two-point grid necessarily chooses an endpoint. These observations do not establish the benefit of a larger grid, and confirmation outcomes are not used to retune any method.

Method	Parameter	Choices	Lower	Upper	Declared grid
Development: Raw ridge	alpha	128	84	4	0.01,1,100
Development: Raw RBF	alpha	128	35	4	0.01,1,100
Development: Affine local	alpha	128	45	8	0.01,1,100
Development: Locality local	alpha	128	35	9	0.01,1,100
Development: KAHM local	alpha	128	51	7	0.01,1,100
Development: Raw + affine	alpha	128	32	4	0.01,1,100
Development: Raw + locality	alpha	128	25	5	0.01,1,100
Development: Raw + KAHM	alpha	128	30	4	0.01,1,100
Development: Histogram boosting	regularization	128	78	50	0.01,1
Development: MLP	regularization	128	26	102	0.01,1
Confirmation: Raw ridge	alpha	80	20	56	0.01,1,100
Confirmation: Raw RBF	alpha	80	13	56	0.01,1,100
Confirmation: Affine local	alpha	80	9	58	0.01,1,100
Confirmation: Locality local	alpha	80	9	60	0.01,1,100
Confirmation: KAHM local	alpha	80	8	59	0.01,1,100
Confirmation: Raw + affine	alpha	80	12	59	0.01,1,100
Confirmation: Raw + locality	alpha	80	12	61	0.01,1,100
Confirmation: Raw + KAHM	alpha	80	12	60	0.01,1,100
Confirmation: Histogram boosting	regularization	80	34	46	0.01,1
Confirmation: MLP	regularization	80	5	75	0.01,1

H Reconstructing the numerical evidence

The supplement has separate controlled and natural forecasting entry points, plus an integrated review manifest. Its README gives commands to verify the packaged bytes, independently reconstruct archived losses, replay diagnostics, regenerate paper assets, and compile the anonymous manuscript. The scientific implementations and original outcome records remain unchanged. Documentation and review manifests added after execution are labeled as such; they are not new pre-outcome registrations or external timing attestations.

For the controlled study, full regeneration generates new working files from the archived configurations and retained seed streams, while compact-record auditing checks the existing outcomes. These operations reproduce the same evidence rather than add independent repetitions. The natural study retains both stage freezes, all 208 completed runs, prepared windows, prediction arrays, source snapshot hashes, and original environment/cost records. Its offline audit checks the complete twelve-method roster, targets and forecast origins, all 2,496 losses, and frozen diagnostic outcomes without refitting forecasting models. Replaying a small diagnostic regression can generate different microsecond timing measurements; original measured resource records remain separate from those replay timings.

All eight natural dataset snapshots retain their attribution and licenses. The inherited implementation and template retain required third-party notices. Anonymization omits named study overviews and author-linked locators while preserving the scientific dependency bytes and identifying the scope of the review export. The separately supplied related certificate manuscript is background for the scope comparison, not a source of additional experimental replications or a certificate for the present fitted predictors.

I Bounded sensitivity to geometry scaling

I.1 Execution boundary and estimand

The original experiments in Sections 4 and 5 use training z-score geometry scaling. The sensitivity analysis was specified after inspecting those experiments. Its protocol and execution sources were published and hashed before the sensitivity fits, and both fixed scaler arms were retained. This ordering prevents choosing the sensitivity cells after their outcomes; it does not make the reused natural sources or synthetic draws fresh confirmation evidence. No adoption diagnostic or best-scaler selection rule was refitted.

The intervention replaces only the geometry input scaler by

$$z_j(x) = 2\frac{x_j - m_j}{M_j - m_j} - 1, \qquad m_j = \min_{i\in\mathcal{T}} x_{ij}, \quad M_j = \max_{i\in\mathcal{T}} x_{ij}.$$

The fitting set $\mathcal{T}$ is the exact training input set for that fit, before its geometry subsample. Coordinates with exactly zero training range are mapped to zero for every geometry query, matching the historical implementation. Validation and test queries are not clipped; values can exceed $[-1, 1]$. Raw references and the raw skip retain their original training standardization. Feature standardization, target standardization, loss scales, forecast horizons, data and model seeds, hyperparameter grids, query arithmetic, and the controlled study's known-support output clipping are unchanged.

Changing the scaler inside the geometry pipeline matters. It can alter partitions, truncated principal directions, absolute reconstruction distances, and angles relative to the origin. The historical kernel fit also uses a regularization level depending on the squared reference-coordinate magnitude. Downstream feature standardization does not reverse these nonlinear changes. By contrast, a nonconstant coordinate's positive affine min–max transform followed by refitted z-score standardization cancels in exact arithmetic when threshold-based scale fallbacks are inactive: $[(ax+b)-(a\mu+b)]/(a\sigma) = (x-\mu)/\sigma$ for $a > 0$. The experiment therefore replaces the geometry scaler itself.

I.2 Natural-source execution

All 208 original configurations are reused: 128 on the original development sources and 80 on the four formerly reserved sources. Each configuration runs raw plus affine, raw plus locality, and raw plus KAHM under both scalers, together with unchanged raw ridge and raw RBF references: 1,664 standalone method runs. The local-only natural architectures and the frozen benefit diagnostic are outside this bounded sensitivity. Each fixed arm uses the original validation-only alpha selection; the selected model is then refitted on the original admissible pretest windows. Its scaler is fitted anew on that refit set, without test observations.

The primary descriptive contrast is min–max minus standard raw-plus-KAHM scaled MSE on the four formerly reserved source families. Each mean uses the original source/series/regime/capacity/seed hierarchy. The original development sources are reported separately. Source-wise and leave-one-source-out results describe heterogeneity; neither seeds nor overlapping histories supply additional independent source families. There are no new confirmatory p-values or simultaneous superiority claims.

Methods execute serially in one benchmark worker with one BLAS thread; method order is shuffled deterministically by task identity, independently of outcomes. New warm-library costs include alpha search, selection cleanup, selected refit, prediction, and serialization. Current data loading, launch-to-result wall time, and the additional sensitivity diagnostics are recorded separately; the original source preparation is reused, rather than freshly timed. The launch-to-result field includes child execution and communication, rather than isolating startup overhead. The inherited import timer starts after some adapter dependencies

Method	Standard MSE	Min–max MSE	Change (%)
Raw ridge	1.1769	1.1769	fixed
Raw RBF	3.7827	3.7827	fixed
Raw + affine	1.4475	1.3306	-8.08
Raw + locality	3.6030	3.3309	-7.55
Raw + KAHM	2.1768	2.3444	+7.70

Table 34: The same fixed sensitivity comparison on the four original development sources. Their outcomes remain separate from the formerly reserved sources. Scaled MSE uses the original equal-source hierarchy; change is min–max versus standard in percent. Raw references are fixed, and no scaler is selected from these test outcomes.

have loaded, so it does not measure full cold startup. New cost comparisons use references remeasured in this execution. They must not be formed by dividing a new method time by an old archived baseline time.

I.3 Controlled execution

The sensitivity recreates the exact 24 complete phase-four data/fit draws for each of autoregression, Van der Pol, and Duffing, at 64 and 256 landmarks. All four original representations are retained under both geometry scalers: raw, affine local coordinates, KAHM local coordinates, and KAHM scores. An explicitly added locality-coordinate control uses the historical native-affine locality gates with the same local principal coordinates. Thus 72 paired jobs produce 1,440 head fits. The raw representation is identical across scaler arms. The original target bounds and validation selection rules remain in force.

For each law/capacity cell, the unit of paired dispersion is a complete data/fit draw. Episode losses and horizon errors are retained but are not counted as additional independent fitted-method repetitions. The analysis reports means and paired standard deviations/standard errors descriptively; the original contrast families' confidence statements are not extended to these post-hoc comparisons. Controlled banks are shared across their representations, so controlled execution times are not standalone deployment cost estimates.

I.4 Numerical verification and retained evidence

The standard natural arms and fixed references are checked against all 1,040 corresponding task–method prediction arrays and selected hyperparameters. All 576 original standard controlled method/capacity fits are checked against archived predictions, losses, and selected hyperparameters; 144 controlled raw cross-scaler pairs are checked for equality. The separate audit reconstructs the new natural losses from per-origin forecasts and all controlled losses from the full saved prediction arrays before compact export.

The review supplement retains every natural sensitivity forecast and all controlled per-episode losses, execution records, selected settings, scaler statistics, out-of-range diagnostics, and file digests. Full controlled predictions and fitted banks are regenerated by the supplied runner; their pre-export verification is recorded in the retained full-prediction audit. Replaying the compact controlled losses is distinguished from independently recreating and checking those full prediction arrays. The original studies' scientific sources, freezes, and outcomes remain unchanged.

I.5 Natural heterogeneity and costs

Table 34 keeps the original development outcomes separate. KAHM's development loss increases by 7.70%; the source changes are -1.99% on Electricity, $+2.06\%$ on NN5, -5.20% on Tourism, and $+19.54\%$ on FRED-MD. Removing FRED-MD changes the aggregate scaler effect to a 3.62% reduction. Neither development aggregation should be used to select a scaler on those already inspected outcomes. Table 35 reports both original role groups, and Tables 36 and 37 retain every fixed method.

Source	K_s	K_m	$\Delta\%$	R	A_m	L_m
Original development sources						
Electricity	0.4539	0.4448	-1.99	0.4388	0.4584	0.4387
NN5	0.7704	0.7863	+2.06	0.7552	0.7824	0.7516
Tourism	3.2286	3.0606	-5.20	3.4759	3.0768	3.0986
FRED-MD	4.2545	5.0857	+19.54	10.4610	1.0050	9.0346
Formerly reserved sources						
Traffic	0.5768	0.5759	-0.15	0.5739	0.5750	0.5722
Hospital	0.9238	0.9158	-0.86	0.8228	0.9157	0.8371
Saugeen	0.6012	0.6067	+0.90	0.6106	0.6646	0.6773
US births	1.2659	1.2704	+0.35	1.2228	1.2830	1.3085

Table 35: Source-level scaling sensitivity. K_s and K_m denote raw + KAHM with standard and min–max geometry; R, A_m, and L_m denote raw RBF, raw + min–max affine, and raw + min–max locality. Entries are original-hierarchy scaled MSE; $\Delta\% = 100(K_m/K_s - 1)$. Original role labels identify the historical design, while every newly computed comparison is post-hoc.

Method / geometry scaler	Electricity	NN5	Tourism	FRED-MD
Raw ridge	0.4243	0.8584	2.8648	0.5601
Raw RBF	0.4388	0.7552	3.4759	10.4610
Raw + affine / std.	0.4481	0.7614	3.1717	1.4090
Raw + affine / min–max	0.4584	0.7824	3.0768	1.0050
Raw + locality / std.	0.4418	0.7474	3.0458	10.1769
Raw + locality / min–max	0.4387	0.7516	3.0986	9.0346
Raw + KAHM / std.	0.4539	0.7704	3.2286	4.2545
Raw + KAHM / min–max	0.4448	0.7863	3.0606	5.0857

Table 36: Every fixed sensitivity method on the original development sources. Entries are source-level scaled MSE using the original nested task weights. The two raw references retain their original standardization.

Method / geometry scaler	Traffic	Hospital	Saugeen	US births
Raw ridge	0.5737	0.8461	0.5220	1.1775
Raw RBF	0.5739	0.8228	0.6106	1.2228
Raw + affine / std.	0.5764	0.9198	0.6068	1.2802
Raw + affine / min–max	0.5750	0.9157	0.6646	1.2830
Raw + locality / std.	0.5720	0.9356	0.6002	1.2770
Raw + locality / min–max	0.5722	0.8371	0.6773	1.3085
Raw + KAHM / std.	0.5768	0.9238	0.6012	1.2659
Raw + KAHM / min–max	0.5759	0.9158	0.6067	1.2704

Table 37: Every fixed sensitivity method on the formerly reserved sources. Entries are source-level scaled MSE using the original nested task weights. The two raw references retain their original standardization.

The source-balanced fraction of test input coordinates outside the final refit range is 3.77% on the formerly reserved sources and 21.45% on the original development sources. The corresponding fractions of histories with at least one such coordinate are 33.08% and 49.26%. The largest absolute transformed test coordinate is 3.1990 and 1.8638, respectively. No final natural refit has a constant input coordinate. These diagnostics confirm that unclipped min–max scaling is distinct from enforcing bounded query inputs; a clipped alternative was not tested.

Table 38 gives the new warm method stages. The retained KAHM min–max Saugeen configuration with history 28, horizon 7, medium capacity, and seed 17 takes 6.2914 seconds, including 4.7618 seconds of search and 1.3632 seconds of cleanup. It is included at its original hierarchical weight, without trimming or a timing-based rerun. The single execution cannot distinguish algorithmic cost changes from runtime variation with the precision of a separate repeated timing benchmark.

Method / geometry scaler	Search	Cleanup	Refit	Forecast	Serialize	Total
Raw ridge	0.0065	0.1770	0.0022	0.0001	0.0005	0.1863
Raw RBF	0.0136	0.1557	0.0062	0.0003	0.0005	0.1762
Raw + affine / std.	0.2316	0.1554	0.0530	0.0036	0.0021	0.4457
Raw + affine / min–max	0.0909	0.1639	0.0593	0.0025	0.0012	0.3178
Raw + locality / std.	0.0887	0.1671	0.0612	0.0022	0.0018	0.3210
Raw + locality / min–max	0.0825	0.1557	0.0672	0.0025	0.0029	0.3108
Raw + KAHM / std.	0.1162	0.1594	0.0939	0.0094	0.0015	0.3804
Raw + KAHM / min–max	0.2682	0.1934	0.0845	0.0084	0.0016	0.5561

Table 38: Contemporaneous, source-balanced warm method times (seconds) on the four formerly reserved sources. Each method independently performs its alpha search, selected refit, prediction, serialization, and cleanup. These totals exclude reused data preparation, current input loading, process startup, and additional sensitivity audits. Launch-to-result and audit elapsed times are recorded separately; the inherited partial import timer is not a complete cold-start measure. Every timing, including outliers, is retained. These single-execution observations are implementation-specific and should not be divided by timings from the earlier execution.

I.6 Complete controlled sensitivity

Table 39 includes all ten representation/scaler arms in every law/capacity cell. KAHM local-coordinate mean error changes by $-0.24/-0.32\%$ on autoregression, $-0.18/+0.23\%$ on Van der Pol, and $-2.85/-7.35\%$ on Duffing at 64/256 landmarks. The main mean rankings are unchanged. The score-only KAHM representation is more sensitive on Van der Pol, where min–max raises its error by 6.03% and 4.78%. A secondary ordering consequently reverses at 256 landmarks: standardized KAHM scores have mean MSE 1.0499×10^{-3}, below affine-local 1.0780×10^{-3}, whereas their min–max errors are 1.1001×10^{-3} and 1.0812×10^{-3}, respectively. The persistence claim concerns the main KAHM-local versus raw/control comparisons, not every pair of methods. The retained repetition-level CSVs supply all paired differences and standard errors; the displayed standard deviations describe variability and do not imply a new simultaneous confidence guarantee.

Table 39: Every controlled sensitivity arm, including raw and KAHM scores. Each entry is mean MSE with between-repetition SD in parentheses, multiplied by 10^3, over the same 24 complete draws. Δ is paired min–max-minus-standard mean difference; its parentheses contain paired SD. These are descriptive dispersions on reused draws, not confidence intervals. The raw arms are identical.

Law	M	Features	Standard (SD)	Min–max (SD)	Δ (paired SD)
Nonlinear AR	64	Raw	4.4238 (0.2161)	4.4238 (0.2161)	+0.0000 (0.0000)
Nonlinear AR	64	Affine local	4.9120 (0.3059)	4.8940 (0.2749)	-0.0180 (0.1507)
Nonlinear AR	64	Locality local	4.9306 (0.3008)	4.9414 (0.2705)	+0.0109 (0.1362)
Nonlinear AR	64	KAHM local	4.8795 (0.2769)	4.8678 (0.2601)	-0.0116 (0.0929)
Nonlinear AR	64	KAHM scores	4.7984 (0.2734)	4.8042 (0.2862)	+0.0058 (0.1198)
Nonlinear AR	256	Raw	4.6958 (0.3312)	4.6958 (0.3312)	+0.0000 (0.0000)
Nonlinear AR	256	Affine local	5.2347 (0.3501)	5.1807 (0.4489)	-0.0540 (0.2677)
Nonlinear AR	256	Locality local	5.2540 (0.3511)	5.2577 (0.3890)	+0.0036 (0.1347)
Nonlinear AR	256	KAHM local	5.1749 (0.3303)	5.1584 (0.3832)	-0.0165 (0.1586)
Nonlinear AR	256	KAHM scores	4.8256 (0.2710)	4.8538 (0.3025)	+0.0282 (0.1257)
Van der Pol	64	Raw	2.8586 (0.1574)	2.8586 (0.1574)	+0.0000 (0.0000)
Van der Pol	64	Affine local	1.3767 (0.1578)	1.3482 (0.1225)	-0.0286 (0.0855)
Van der Pol	64	Locality local	1.1324 (0.0737)	1.1240 (0.0805)	-0.0084 (0.0363)
Van der Pol	64	KAHM local	1.0266 (0.0656)	1.0248 (0.0678)	-0.0018 (0.0198)
Van der Pol	64	KAHM scores	1.5711 (0.2109)	1.6658 (0.2221)	+0.0947 (0.1891)
Van der Pol	256	Raw	1.6541 (0.0402)	1.6541 (0.0402)	+0.0000 (0.0000)
Van der Pol	256	Affine local	1.0780 (0.0501)	1.0812 (0.0525)	+0.0032 (0.0297)
Van der Pol	256	Locality local	0.9413 (0.0363)	0.9448 (0.0391)	+0.0036 (0.0191)
Van der Pol	256	KAHM local	0.8720 (0.0293)	0.8740 (0.0277)	+0.0020 (0.0060)
Van der Pol	256	KAHM scores	1.0499 (0.0600)	1.1001 (0.0794)	+0.0502 (0.0735)
Duffing	64	Raw	3.4597 (0.1516)	3.4597 (0.1516)	+0.0000 (0.0000)
Duffing	64	Affine local	3.0549 (0.1764)	2.9981 (0.1357)	-0.0568 (0.1262)
Duffing	64	Locality local	2.8016 (0.1346)	2.8227 (0.1447)	+0.0211 (0.1128)
Duffing	64	KAHM local	4.2154 (0.4775)	4.0951 (0.4723)	-0.1203 (0.4225)
Duffing	64	KAHM scores	4.7005 (0.5431)	4.5547 (0.3330)	-0.1458 (0.4739)
Duffing	256	Raw	2.8364 (0.1029)	2.8364 (0.1029)	+0.0000 (0.0000)
Duffing	256	Affine local	2.7981 (0.1245)	2.7742 (0.1318)	-0.0238 (0.1180)
Duffing	256	Locality local	2.7006 (0.1398)	2.6921 (0.1321)	-0.0085 (0.1001)
Duffing	256	KAHM local	4.0736 (0.4520)	3.7741 (0.3794)	-0.2995 (0.3555)
Duffing	256	KAHM scores	3.3808 (0.2940)	3.3354 (0.2419)	-0.0454 (0.2415)